

abs_4

September 10, 2025

```
[38]: import sys
import os
sys.path.append(os.path.abspath("../src"))
```

```
[39]: DEBUG = True

from termcolor import cprint

def debug_info(*args):
    """Prints a yellow 'INFO' message if DEBUG is True."""
    if DEBUG:
        cprint(f"[INFO] {' '.join(map(str, args))}", "yellow")

def debug_error(*args):
    """Prints a red 'ERROR' message if DEBUG is True."""
    if DEBUG:
        cprint(f"[ERROR] {' '.join(map(str, args))}", "red")

def debug_success(*args):
    """Prints a green 'SUCCESS' message if DEBUG is True."""
    if DEBUG:
        cprint(f"[SUCCESS] {' '.join(map(str, args))}", "green")
```

```
[40]: from pathlib import Path

def get_dataset_directory():
    current_path = Path.cwd()
    base_project_dir = current_path.parent
    dataset_directory = base_project_dir / "src" / "abstractionssymh" / 
    ↪ "dataset"

    debug_info("Current notebook location:", current_path)
    debug_info("Base project directory:", base_project_dir)
    debug_info("Target dataset directory:", dataset_directory)

    return dataset_directory

dataset_directory = get_dataset_directory()
```

[INFO] Current notebook location: /Users/amogh/Projects/abstraction-
discovery/notebookssymh
[INFO] Base project directory: /Users/amogh/Projects/abstraction-
discovery
[INFO] Target dataset directory: /Users/amogh/Projects/abstraction-
discovery/src/abstractionssymh/dataset

```
[41]: import random
from abstractionssymh.data_loader import parse_json_to_dsl

def load_chair_dsl(chair_directory, use_random=False):
    json_files = sorted(chair_directory.glob("*.json"))
    if not json_files:
        debug_error("No JSON files found in:", chair_directory)
        return None

    file_path = random.choice(json_files) if use_random else json_files[0]
    debug_success("Loading chair file:", file_path)

    json_content = file_path.read_text(encoding="utf-8")
    dsl_object = parse_json_to_dsl(json_content)

    debug_success("Successfully parsed DSL object")
    debug_info(dsl_object)
    return dsl_object

chair_directory = dataset_directory / "Chair"
dsl_object = load_chair_dsl(chair_directory, use_random=False)
```

[SUCCESS] Loading chair file: /Users/amogh/Projects/abstraction-
discovery/src/abstractionssymh/dataset/Chair/Chair_1.json
[SUCCESS] Successfully parsed DSL object

```

[INFO] Union(
  Union(
    Translate(center=[0.003, 0.149, 0.072])
      Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
        Scale(lengths=[0.891, 0.224, 0.806])
          Box(label=1),
    Translate(center=[0.004, -0.422, 0.170])
      Rotate(quat=[-0.3389, -0.0000, 0.0065, 0.9408])
        Scale(lengths=[0.898, 0.524, 1.154])
          Box(label=2)
  ),
  Translate(center=[0.030, 0.488, -0.315])
    Rotate(quat=[-0.0821, -0.0071, -0.0494, 0.9954])
      Scale(lengths=[0.892, 0.781, 0.305])
        Box(label=0)
)

```

```
[42]: from abstractionssymh.plot_utils import plot_dsl_with_k3d
```

```

def plot_chair(dsl_obj):
    if dsl_obj is None:
        debug_error("No DSL object to plot.")
        return
    try:
        debug_info("Rendering DSL object with k3d...")
        plot_dsl_with_k3d(dsl_obj)
        debug_success("Plotting complete.")
    except Exception as e:
        debug_error("Failed to plot DSL object:", e)

plot_chair(dsl_object)

```

```
[INFO] Rendering DSL object with k3d...
```

```
Expanding DSL tree for visualization...
```

```
Found 3 total boxes after expansion.
```

```
Output()
```

```
[SUCCESS] Plotting complete.
```

```
[43]: from collections import defaultdict

def find_all_subtrees(node):
    """Recursively yields all subtrees (nodes) in a DSL object."""
    name, (params, children) = type(node).__name__, node.serialize()[1]
    for child in children:
        if hasattr(child, "serialize"):
            yield from find_all_subtrees(child)
    yield node

[44]: def collect_singleton_and_pair_data(dsl_shapes):
    """Collects singleton parameters and parent-child pair parameters."""
    if not dsl_shapes:
        debug_error("No DSL shapes provided for analysis.")
        return {}, {}

    s_data, p_data = defaultdict(list), defaultdict(list)

    for shape in dsl_shapes:
        for node in find_all_subtrees(shape):
            name, (p_params, children) = type(node).__name__, node.
↪serialize()[1]

            if p_params:
                s_data[name].append(p_params)

            for child in (c for c in children if hasattr(c, "serialize")):
                c_name, (c_params, _) = type(child).__name__, child.
↪serialize()[1]

                combo_params = p_params + c_params
                if combo_params:
                    pair_sig = f"{name}({c_name})"
                    p_data[pair_sig].append(combo_params)

    debug_success("Collected keys:",
                  f"{s_data.keys()} singletons, {p_data.keys()} pairs")
    debug_success("Collected parameters:",
                  f"{len(s_data)} singletons, {len(p_data)} pairs")
    return dict(sorted(s_data.items())), dict(sorted(p_data.items()))

[45]: import torch
import torch.nn as nn
from torch.utils.data import DataLoader, TensorDataset
from torch.optim import AdamW
from abstractionssymh.dsl_nodes import Box, Scale, Rotate, Translate, Union,
↪SymRef, SymRot, SymTrans
```

```

class Autoencoder(nn.Module):
    """Simple fully-connected autoencoder with configurable hidden dim."""
    def __init__(self, input_dim: int, hidden_dim: int):
        super().__init__()
        debug_info(f"Initializing Autoencoder: input_dim={input_dim},
↳hidden_dim={hidden_dim}")

        self.encoder = nn.Sequential(
            nn.Linear(input_dim, 32), nn.ReLU(),
            nn.Linear(32, 16), nn.ReLU(),
            nn.Linear(16, hidden_dim)
        )
        self.decoder = nn.Sequential(
            nn.Linear(hidden_dim, 16), nn.ReLU(),
            nn.Linear(16, 32), nn.ReLU(),
            nn.Linear(32, input_dim)
        )

        debug_success("Autoencoder initialized successfully.")

    def forward(self, x):
        # debug_info(f"Forward pass: input shape={tuple(x.shape)}")
        latent = self.encoder(x)
        # debug_info(f"Encoded latent shape={tuple(latent.shape)}")
        decoded = self.decoder(latent)
        # debug_info(f"Decoded output shape={tuple(decoded.shape)}")
        return latent, decoded

```

```

[46]: def instantiate_pattern(pattern_name, params, children):
    """
    Rebuilds a DSL sub-tree from its pattern name, parameters, and preserved_
↳children.
    Handles both SINGLETON and PAIR patterns.
    """
    debug_info(f"[INSTANTIATE] Pattern: '{pattern_name}' | params={params[:
↳10]}{'...' if len(params)>10 else ''} | #children={len(children)}")

    # Registry defining parent parameter lengths for known PAIRS
    param_split = {
        'Scale(Box)': 3, 'Rotate(Scale)': 4, 'Translate(Rotate)': 3,
        'Union(Translate)': 0, 'Union(SymRef)': 0, 'Union(SymRot)': 0,
↳'Union(SymTrans)': 0,
        'SymRef(Union)': 6, 'SymRef(Translate)': 6, 'SymRot(Union)': 6,
        'SymRot(Translate)': 6, 'SymTrans(Translate)': 3
    }

```

```

# --- SINGLETON RECONSTRUCTION ---
if pattern_name in ['Scale', 'Rotate', 'Translate', 'SymRef', 'SymRot',
↪ 'SymTrans']:
    NodeClass = globals()[pattern_name]
    child_node = children[0] if children else Box(-1)

    if NodeClass == SymRef:
        debug_info(f" [SINGLETON] SymRef with params length {len(params)}")
        return SymRef(child_node, plane_normal=params[:3],
↪ point_on_plane=params[3:])
    if NodeClass == SymRot:
        debug_info(f" [SINGLETON] SymRot with params length {len(params)}")
        return SymRot(child_node, axis=params[:3], center=params[3:],
↪ n_fold=-1)
    if NodeClass == SymTrans:
        debug_info(f" [SINGLETON] SymTrans with params length
↪ {len(params)}")
        return SymTrans(child_node, end_point=params, n_fold=-1)

    return NodeClass(child_node, params)

# --- PAIR RECONSTRUCTION ---
elif pattern_name in param_split:
    p_len = param_split[pattern_name]
    p_params, c_params = params[:p_len], params[p_len:]
    parent_name, child_name = pattern_name.split('(')[0], pattern_name.
↪ split('(')[1][:-1]
    ParentClass, ChildClass = globals()[parent_name], globals()[child_name]

    # --- Build child node ---
    grandchild_node = children[0] if children else Box(-1)
    if ChildClass == Box:
        child_node = Box(-1)
    elif ChildClass == Union:
        if len(children) >= 2:
            child_node = Union(children[0], children[1])
        else:
            debug_error(f" [PAIR] Unexpected children for Union in
↪ {pattern_name}")
            child_node = Box(-1)
    elif ChildClass == SymRef:
        child_node = SymRef(grandchild_node, plane_normal=c_params[:3],
↪ point_on_plane=c_params[3:])
    elif ChildClass == SymRot:
        child_node = SymRot(grandchild_node, axis=c_params[:3],
↪ center=c_params[3:], n_fold=-1)

```

```

        elif ChildClass == SymTrans:
            child_node = SymTrans(grandchild_node, end_point=c_params,
↪n_fold=-1)
        else:
            child_node = ChildClass(grandchild_node, c_params)

        # --- Build parent node ---
        if ParentClass == Union:
            if len(children) >= 2:
                return Union(child_node, children[1])
            else:
                debug_error(f" [PAIR] Union parent with missing second child,
↪in {pattern_name}")
                return Box(-1)
        elif ParentClass == SymRef:
            return SymRef(child_node, plane_normal=p_params[:3],
↪point_on_plane=p_params[3:])
        elif ParentClass == SymRot:
            return SymRot(child_node, axis=p_params[:3], center=p_params[3:],
↪n_fold=-1)
        elif ParentClass == SymTrans:
            return SymTrans(child_node, end_point=p_params, n_fold=-1)
        else:
            return ParentClass(child_node, p_params)

    debug_error(f"[UNKNOWN PATTERN] '{pattern_name}', defaulting to Box(-1)")
    return Box(-1)

```

[47]: `import textwrap`

```

class Abstraction:
    def __init__(self, pattern_name, compressed_params, model, children=None):
        self.pattern_name, self.compressed_params, self.model = pattern_name,
↪compressed_params, model
        self.children = children if children is not None else []

    def __str__(self):
        # Create the header for the Abstraction node
        header = f"Abs({self.pattern_name}, dim={len(self.compressed_params)})"

        # If the abstraction has preserved children, indent and print them
        if not self.children:
            return header
        else:
            # Indent each child's own string representation

```

```

        child_strs = [textwrap.indent(str(c), '    ') for c in self.
↪children]
        # Join the indented child strings
        return f"{header}(\n" + ",\n".join(child_strs) + "\n)"

    __repr__ = __str__    # makes debugging in lists/dicts nicer

    def expand(self):
        """Reconstructs the full DSL node from compressed parameters."""
        debug_info(f"Expanding abstraction: {self}")

        if not self.compressed_params:
            debug_error("No compressed params found. Returning Box(-1).")
            return Box(-1)    # fallback if params are missing

        self.model.eval()
        with torch.no_grad():
            params_tensor = torch.tensor(
                self.compressed_params, dtype=torch.float32
            ).unsqueeze(0)
            debug_info(f"Compressed params tensor shape: {tuple(params_tensor.
↪shape))}")
            reconstructed_params = self.model.decoder(params_tensor).squeeze().
↪tolist()
            debug_success(f"Reconstructed params: {reconstructed_params}")

            rebuilt_node = instantiate_pattern(self.pattern_name,
↪reconstructed_params, self.children)
            debug_success(f"Successfully rebuilt node: {rebuilt_node}")
            return rebuilt_node.expand()

```

```

[48]: def prepare_autoencoder_train_data(parameters, mask, batch_size=64):
        """Creates a DataLoader from parameters and a boolean mask."""
        debug_info(f"Preparing DataLoader: batch_size={batch_size}, mask sum={mask.
↪sum().item()}")
        tensor = torch.tensor(parameters, dtype=torch.float32)
        if mask.shape[0] != tensor.shape[0]:
            debug_error(f"Mask size {mask.shape[0]} does not match data size
↪{tensor.shape[0]}")
            raise ValueError(f"Mask size {mask.shape[0]} != data size {tensor.
↪shape[0]}")
        dataloader = DataLoader(TensorDataset(tensor[mask]), batch_size=batch_size,
↪shuffle=True)
        debug_success("DataLoader prepared successfully.")
        return dataloader

```

```
[49]: def is_well_explained(model, parameters_tensor, error_threshold):
    """Checks if parameters are well reconstructed by the autoencoder."""
    model.eval()
    with torch.no_grad():
        _, reconstructions = model(parameters_tensor)
        error, _ = torch.max(torch.abs(reconstructions - parameters_tensor),
        ↪dim=-1)
        debug_info(f"Max reconstruction error per sample: {error}")
        well_explained = error < error_threshold
        debug_info(f"Number of well-explained examples: {well_explained.sum().
        ↪item()}/{len(well_explained)}")
        return well_explained

[50]: def train_autoencoder(model, dataloader, epochs=50, lr=1e-3):
    """Trains an autoencoder on the given DataLoader."""
    debug_info(f"Training autoencoder for {epochs} epochs, learning rate={lr}")
    optimizer = AdamW(model.parameters(), lr=lr)
    loss_fn = nn.MSELoss()
    model.train()

    for epoch in range(epochs):
        epoch_loss = 0.0
        for batch in dataloader:
            x = batch[0]
            optimizer.zero_grad()
            _, x_rec = model(x)
            loss = loss_fn(x_rec, x)
            loss.backward()
            optimizer.step()
            epoch_loss += loss.item() * x.size(0)
        debug_info(f"Epoch {epoch+1}/{epochs} - Average loss: {epoch_loss /
        ↪len(dataloader.dataset):.6f}")

    debug_success("Autoencoder training complete.")
    return model

[51]: def find_abstractions(
    structures,
    structure_type="PATTERNS",
    min_examples=100,
    retrain_iterations=3,
    error_threshold=0.05,
    epochs=50,
):
    """Trains autoencoders for each structure type and returns models."""
    trained_models = {}
```

```

sorted_structures = sorted(structures.items(), key=lambda item:
↪len(item[1]), reverse=True)

debug_info(f"Finding abstractions for {len(sorted_structures)} structures
↪of type {structure_type}")

for name, parameters in sorted_structures:
    if len(parameters) < min_examples:
        debug_info(f"Skipping {name} (only {len(parameters)} examples)")
        continue

    num_params = len(parameters[0])
    if num_params <= 1:
        debug_info(f"Skipping {name} (too few params: {num_params})")
        continue

    debug_success(f"Training model for {name} with {len(parameters)}
↪samples, {num_params} params")
    params_tensor = torch.tensor(parameters, dtype=torch.float32)
    mask = torch.ones(len(parameters), dtype=torch.bool)

    for iteration in range(retrain_iterations):
        if not mask.any():
            debug_info(f"All {name} examples filtered out at iteration
↪{iteration}")
            break

        debug_info(f"Iteration {iteration+1}/{retrain_iterations} for
↪{name}")
        dataloader = prepare_autoencoder_train_data(parameters, mask=mask)
        model = Autoencoder(num_params, max(1, num_params - 1))
        model = train_autoencoder(model, dataloader, epochs=epochs)

        mask = is_well_explained(model, params_tensor, error_threshold)

    trained_models[name] = model
    debug_success(f"Model training finished for {name}. Total
↪well-explained examples: {mask.sum().item()}/{len(parameters)}")

    debug_success(f"Finished training {len(trained_models)} {structure_type}")
    return trained_models

```

```

[52]: def integrate_abstractions(node, singleton_models, pair_models,
↪error_threshold, depth=0):
    indent = "  " * depth

```

```

    if isinstance(node, Abstraction):
        debug_info(f"{indent}[Abstraction] Node already abstracted: {node.
↪pattern_name}")
        return node

    # 1. Recurse to children first
    _, (_, children) = node.serialize()
    rebuilt_children = [
        integrate_abstractions(c, singleton_models, pair_models,
↪error_threshold, depth+1)
        if hasattr(c, 'serialize') else c
        for c in children
    ]

    # 2. Rebuild current node
    try:
        if isinstance(node, Union):
            current_node = Union(rebuilt_children[0], rebuilt_children[1])
        elif isinstance(node, SymRef):
            current_node = SymRef(rebuilt_children[0], plane_normal=node.plane,
↪point_on_plane=node.point_on_plane)
        elif isinstance(node, SymRot):
            current_node = SymRot(rebuilt_children[0], axis=node.axis,
↪center=node.center, n_fold=node.n)
        elif isinstance(node, SymTrans):
            current_node = SymTrans(rebuilt_children[0], end_point=node.
↪end_point, n_fold=node.n)
        elif hasattr(node, 'child'):
            kwargs = {k:v for k,v in node.__dict__.items() if k != 'child'}
            current_node = type(node)(rebuilt_children[0], **kwargs)
        else:
            current_node = type(node)(*rebuilt_children)
            debug_info(f"{indent}[Rebuilt] {type(current_node).__name__} with
↪{len(rebuilt_children)} children")
        except Exception as e:
            debug_error(f"{indent}[Rebuild FAILED] {type(node).__name__} at depth
↪{depth}: {e}")
        return node

    # 3. Try pair abstraction
    child_nodes = [c for c in rebuilt_children if hasattr(c, 'serialize')]
    if len(child_nodes) == 1:
        child_node = child_nodes[0]
        if not isinstance(child_node, Abstraction):
            pair_sig = f"{type(current_node).__name__}({type(child_node).
↪__name__})"

```

```

        if pair_sig in pair_models:
            model, (p_params, _), (c_params, c_children) = pair_models[pair_sig], current_node.serialize()[1], child_node.serialize()[1]
            if p_params + c_params:
                combined = torch.tensor(p_params + c_params, dtype=torch.float32).unsqueeze(0)
                _, reconstruction = model(combined)
                error = torch.max(torch.abs(reconstruction - combined)).item()
                debug_info(f"{indent}[PAIR CHECK] {pair_sig}, error={error:.4f}")
                if error < error_threshold:
                    encoding, _ = model(combined)
                    grandchildren = [c for c in c_children if hasattr(c, 'serialize')]
                    debug_success(f"{indent}[PAIR ABSTRACTION CREATED] {pair_sig}")
                    return Abstraction(pair_sig, encoding.squeeze().tolist(), model, children=grandchildren)

    # 4. Try singleton abstraction
    name = type(current_node).__name__
    if name in singleton_models:
        model, (p_params, _) = singleton_models[name], current_node.serialize()[1]
        if p_params:
            params_tensor = torch.tensor(p_params, dtype=torch.float32).unsqueeze(0)
            _, reconstruction = model(params_tensor)
            error = torch.max(torch.abs(reconstruction - params_tensor)).item()
            debug_info(f"{indent}[SINGLETON CHECK] {name}, error={error:.4f}")
            if error < error_threshold:
                encoding, _ = model(params_tensor)
                debug_success(f"{indent}[SINGLETON ABSTRACTION CREATED] {name}")
                return Abstraction(name, encoding.squeeze().tolist(), model, children=child_nodes)

    # 5. No abstraction, return rebuilt node
    debug_info(f"{indent}[RETURN NODE] {type(current_node).__name__}")
    return current_node

```

[53]: # Load DSL shapes

```

chair_directory = dataset_directory / "Chair"
json_files = sorted(list(chair_directory.glob("*.json")))

```

```

try:
    debug_info("Starting to load DSL shapes from JSON files...")
    all_dsl_shapes = [parse_json_to_dsl(Path(f).read_text()) for f in ↵
        ↵json_files[:1000]]
    debug_success(f"Loaded {len(all_dsl_shapes)} shapes from dataset.")
except Exception as e:
    debug_error("Failed to load DSL shapes:", e)
    all_dsl_shapes = []

# Collect parameters
if all_dsl_shapes:
    debug_info("Collecting singleton and pair parameters from loaded shapes...")
    singleton_params, pair_params = ↵
    ↵collect_singleton_and_pair_data(all_dsl_shapes)

    num_singleton = sum(len(v) for v in singleton_params.values())
    num_pair = sum(len(v) for v in pair_params.values())
    debug_success(f"Collected {num_singleton} singleton and {num_pair} pair ↵
        ↵parameter sets.")

    # Report top structures by count
    # top_singletons = sorted(singleton_params.items(), key=lambda x: ↵
    ↵len(x[1]), reverse=True)[:5]
    # top_pairs = sorted(pair_params.items(), key=lambda x: len(x[1]), ↵
    ↵reverse=True)[:5]
    # debug_info("Top singleton structures by example count:", top_singletons)
    # debug_info("Top pair structures by example count:", top_pairs)

    # Train singleton models
    debug_info("Starting singleton model training...")
    singleton_models = find_abstractions(singleton_params, ↵
    ↵structure_type="SINGLETONS", min_examples=50, epochs=25)

    # Train pair models
    debug_info("Starting pair model training...")
    pair_models = find_abstractions(pair_params, structure_type="PAIRS", ↵
    ↵min_examples=50, epochs=25)

    debug_success(f"Training complete. {len(singleton_models)} singleton models ↵
    ↵and {len(pair_models)} pair models trained.")
else:
    debug_error("No DSL shapes available. Abstraction pipeline skipped.")

```

[INFO] Starting to load DSL shapes from JSON files...

[SUCCESS] Loaded 1000 shapes from dataset.

[INFO] Collecting singleton and pair parameters from loaded shapes...

```

[SUCCESS] Collected keys: dict_keys(['Scale', 'Rotate', 'Translate',
'SymRef', 'SymRot', 'SymTrans']) singletons, dict_keys(['Scale(Box)',
'Rotate(Scale)', 'Translate(Rotate)', 'Union(Translate)', 'SymRef(Union)',
'Union(SymRef)', 'SymRef(Translate)', 'SymRot(Union)', 'Union(SymRot)',
'SymRot(Translate)', 'SymTrans(Translate)', 'Union(SymTrans)']) pairs
[SUCCESS] Collected parameters: 6 singletons, 12 pairs
[SUCCESS] Collected 21173 singleton and 28701 pair parameter sets.
[INFO] Starting singleton model training...
[INFO] Finding abstractions for 6 structures of type SINGLETONS
[SUCCESS] Training model for Rotate with 6386 samples, 4 params
[INFO] Iteration 1/3 for Rotate
[INFO] Preparing DataLoader: batch_size=64, mask sum=6386
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=4, hidden_dim=3
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.037681
[INFO] Epoch 2/25 - Average loss: 0.003224
[INFO] Epoch 3/25 - Average loss: 0.002967
[INFO] Epoch 4/25 - Average loss: 0.001811
[INFO] Epoch 5/25 - Average loss: 0.000644
[INFO] Epoch 6/25 - Average loss: 0.000455
[INFO] Epoch 7/25 - Average loss: 0.000428
[INFO] Epoch 8/25 - Average loss: 0.000414
[INFO] Epoch 9/25 - Average loss: 0.000401
[INFO] Epoch 10/25 - Average loss: 0.000371
[INFO] Epoch 11/25 - Average loss: 0.000313
[INFO] Epoch 12/25 - Average loss: 0.000172
[INFO] Epoch 13/25 - Average loss: 0.000063
[INFO] Epoch 14/25 - Average loss: 0.000042
[INFO] Epoch 15/25 - Average loss: 0.000032
[INFO] Epoch 16/25 - Average loss: 0.000024
[INFO] Epoch 17/25 - Average loss: 0.000019
[INFO] Epoch 18/25 - Average loss: 0.000016
[INFO] Epoch 19/25 - Average loss: 0.000014
[INFO] Epoch 20/25 - Average loss: 0.000013
[INFO] Epoch 21/25 - Average loss: 0.000011
[INFO] Epoch 22/25 - Average loss: 0.000010
[INFO] Epoch 23/25 - Average loss: 0.000010
[INFO] Epoch 24/25 - Average loss: 0.000008
[INFO] Epoch 25/25 - Average loss: 0.000007
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0012, 0.0031, 0.0039,
..., 0.0012, 0.0012, 0.0016])
[INFO] Number of well-explained examples: 6378/6386

```

```

[INFO] Iteration 2/3 for Rotate
[INFO] Preparing DataLoader: batch_size=64, mask sum=6378
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=4, hidden_dim=3
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.057876
[INFO] Epoch 2/25 - Average loss: 0.003083
[INFO] Epoch 3/25 - Average loss: 0.002940
[INFO] Epoch 4/25 - Average loss: 0.002596
[INFO] Epoch 5/25 - Average loss: 0.002061
[INFO] Epoch 6/25 - Average loss: 0.001982
[INFO] Epoch 7/25 - Average loss: 0.001907
[INFO] Epoch 8/25 - Average loss: 0.001551
[INFO] Epoch 9/25 - Average loss: 0.000567
[INFO] Epoch 10/25 - Average loss: 0.000429
[INFO] Epoch 11/25 - Average loss: 0.000408
[INFO] Epoch 12/25 - Average loss: 0.000399
[INFO] Epoch 13/25 - Average loss: 0.000386
[INFO] Epoch 14/25 - Average loss: 0.000371
[INFO] Epoch 15/25 - Average loss: 0.000344
[INFO] Epoch 16/25 - Average loss: 0.000278
[INFO] Epoch 17/25 - Average loss: 0.000167
[INFO] Epoch 18/25 - Average loss: 0.000069
[INFO] Epoch 19/25 - Average loss: 0.000033
[INFO] Epoch 20/25 - Average loss: 0.000027
[INFO] Epoch 21/25 - Average loss: 0.000025
[INFO] Epoch 22/25 - Average loss: 0.000023
[INFO] Epoch 23/25 - Average loss: 0.000022
[INFO] Epoch 24/25 - Average loss: 0.000021
[INFO] Epoch 25/25 - Average loss: 0.000020
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0015, 0.0180, 0.0121,
..., 0.0015, 0.0015, 0.0027])
[INFO] Number of well-explained examples: 6356/6386
[INFO] Iteration 3/3 for Rotate
[INFO] Preparing DataLoader: batch_size=64, mask sum=6356
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=4, hidden_dim=3
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.036764
[INFO] Epoch 2/25 - Average loss: 0.003008
[INFO] Epoch 3/25 - Average loss: 0.002516
[INFO] Epoch 4/25 - Average loss: 0.001711
[INFO] Epoch 5/25 - Average loss: 0.001548
[INFO] Epoch 6/25 - Average loss: 0.001497

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[INFO] Epoch 7/25 - Average loss: 0.001475
[INFO] Epoch 8/25 - Average loss: 0.001462
[INFO] Epoch 9/25 - Average loss: 0.001425
[INFO] Epoch 10/25 - Average loss: 0.001331
[INFO] Epoch 11/25 - Average loss: 0.000982
[INFO] Epoch 12/25 - Average loss: 0.000555
[INFO] Epoch 13/25 - Average loss: 0.000430
[INFO] Epoch 14/25 - Average loss: 0.000378
[INFO] Epoch 15/25 - Average loss: 0.000360
[INFO] Epoch 16/25 - Average loss: 0.000350
[INFO] Epoch 17/25 - Average loss: 0.000350
[INFO] Epoch 18/25 - Average loss: 0.000342
[INFO] Epoch 19/25 - Average loss: 0.000342
[INFO] Epoch 20/25 - Average loss: 0.000336
[INFO] Epoch 21/25 - Average loss: 0.000332
[INFO] Epoch 22/25 - Average loss: 0.000320
[INFO] Epoch 23/25 - Average loss: 0.000302
[INFO] Epoch 24/25 - Average loss: 0.000253
[INFO] Epoch 25/25 - Average loss: 0.000157
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0038, 0.0184, 0.0121,
..., 0.0038, 0.0038, 0.0028])
[INFO] Number of well-explained examples: 6227/6386
[SUCCESS] Model training finished for Rotate. Total well-explained
examples: 6227/6386
[SUCCESS] Training model for Scale with 6386 samples, 3 params
[INFO] Iteration 1/3 for Scale
[INFO] Preparing DataLoader: batch_size=64, mask sum=6386
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.188965
[INFO] Epoch 2/25 - Average loss: 0.059559
[INFO] Epoch 3/25 - Average loss: 0.019671
[INFO] Epoch 4/25 - Average loss: 0.012414
[INFO] Epoch 5/25 - Average loss: 0.010181
[INFO] Epoch 6/25 - Average loss: 0.009003
[INFO] Epoch 7/25 - Average loss: 0.008053
[INFO] Epoch 8/25 - Average loss: 0.007581
[INFO] Epoch 9/25 - Average loss: 0.006985
[INFO] Epoch 10/25 - Average loss: 0.006708
[INFO] Epoch 11/25 - Average loss: 0.006369
[INFO] Epoch 12/25 - Average loss: 0.006243
[INFO] Epoch 13/25 - Average loss: 0.006029
[INFO] Epoch 14/25 - Average loss: 0.005927

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[INFO] Epoch 15/25 - Average loss: 0.005717
[INFO] Epoch 16/25 - Average loss: 0.005747
[INFO] Epoch 17/25 - Average loss: 0.005584
[INFO] Epoch 18/25 - Average loss: 0.005559
[INFO] Epoch 19/25 - Average loss: 0.005492
[INFO] Epoch 20/25 - Average loss: 0.005383
[INFO] Epoch 21/25 - Average loss: 0.005302
[INFO] Epoch 22/25 - Average loss: 0.005159
[INFO] Epoch 23/25 - Average loss: 0.005080
[INFO] Epoch 24/25 - Average loss: 0.005095
[INFO] Epoch 25/25 - Average loss: 0.005079
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0464, 0.2266, 0.0724,
..., 0.0225, 0.0225, 0.0136])
[INFO] Number of well-explained examples: 3286/6386
[INFO] Iteration 2/3 for Scale
[INFO] Preparing DataLoader: batch_size=64, mask sum=3286
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.117876
[INFO] Epoch 2/25 - Average loss: 0.073309
[INFO] Epoch 3/25 - Average loss: 0.030327
[INFO] Epoch 4/25 - Average loss: 0.006043
[INFO] Epoch 5/25 - Average loss: 0.002428
[INFO] Epoch 6/25 - Average loss: 0.001212
[INFO] Epoch 7/25 - Average loss: 0.000890
[INFO] Epoch 8/25 - Average loss: 0.000752
[INFO] Epoch 9/25 - Average loss: 0.000687
[INFO] Epoch 10/25 - Average loss: 0.000649
[INFO] Epoch 11/25 - Average loss: 0.000588
[INFO] Epoch 12/25 - Average loss: 0.000543
[INFO] Epoch 13/25 - Average loss: 0.000515
[INFO] Epoch 14/25 - Average loss: 0.000495
[INFO] Epoch 15/25 - Average loss: 0.000473
[INFO] Epoch 16/25 - Average loss: 0.000464
[INFO] Epoch 17/25 - Average loss: 0.000465
[INFO] Epoch 18/25 - Average loss: 0.000438
[INFO] Epoch 19/25 - Average loss: 0.000427
[INFO] Epoch 20/25 - Average loss: 0.000417
[INFO] Epoch 21/25 - Average loss: 0.000410
[INFO] Epoch 22/25 - Average loss: 0.000409
[INFO] Epoch 23/25 - Average loss: 0.000399
[INFO] Epoch 24/25 - Average loss: 0.000385
[INFO] Epoch 25/25 - Average loss: 0.000388
[SUCCESS] Autoencoder training complete.

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[INFO] Max reconstruction error per sample: tensor([0.0069, 0.2244, 0.0662,
..., 0.0151, 0.0131, 0.0167])
[INFO] Number of well-explained examples: 4035/6386
[INFO] Iteration 3/3 for Scale
[INFO] Preparing DataLoader: batch_size=64, mask sum=4035
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.108484
[INFO] Epoch 2/25 - Average loss: 0.072313
[INFO] Epoch 3/25 - Average loss: 0.026972
[INFO] Epoch 4/25 - Average loss: 0.011607
[INFO] Epoch 5/25 - Average loss: 0.006764
[INFO] Epoch 6/25 - Average loss: 0.002681
[INFO] Epoch 7/25 - Average loss: 0.001023
[INFO] Epoch 8/25 - Average loss: 0.000640
[INFO] Epoch 9/25 - Average loss: 0.000553
[INFO] Epoch 10/25 - Average loss: 0.000498
[INFO] Epoch 11/25 - Average loss: 0.000467
[INFO] Epoch 12/25 - Average loss: 0.000436
[INFO] Epoch 13/25 - Average loss: 0.000400
[INFO] Epoch 14/25 - Average loss: 0.000391
[INFO] Epoch 15/25 - Average loss: 0.000368
[INFO] Epoch 16/25 - Average loss: 0.000361
[INFO] Epoch 17/25 - Average loss: 0.000352
[INFO] Epoch 18/25 - Average loss: 0.000337
[INFO] Epoch 19/25 - Average loss: 0.000333
[INFO] Epoch 20/25 - Average loss: 0.000320
[INFO] Epoch 21/25 - Average loss: 0.000324
[INFO] Epoch 22/25 - Average loss: 0.000310
[INFO] Epoch 23/25 - Average loss: 0.000311
[INFO] Epoch 24/25 - Average loss: 0.000302
[INFO] Epoch 25/25 - Average loss: 0.000295
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0195, 0.1997, 0.0789,
..., 0.0032, 0.0093, 0.0220])
[INFO] Number of well-explained examples: 4255/6386
[SUCCESS] Model training finished for Scale. Total well-explained examples:
4255/6386
[SUCCESS] Training model for Translate with 6386 samples, 3 params
[INFO] Iteration 1/3 for Translate
[INFO] Preparing DataLoader: batch_size=64, mask sum=6386
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.

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[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.072463
[INFO] Epoch 2/25 - Average loss: 0.022042
[INFO] Epoch 3/25 - Average loss: 0.016494
[INFO] Epoch 4/25 - Average loss: 0.012278
[INFO] Epoch 5/25 - Average loss: 0.009312
[INFO] Epoch 6/25 - Average loss: 0.008147
[INFO] Epoch 7/25 - Average loss: 0.007315
[INFO] Epoch 8/25 - Average loss: 0.006883
[INFO] Epoch 9/25 - Average loss: 0.006535
[INFO] Epoch 10/25 - Average loss: 0.006381
[INFO] Epoch 11/25 - Average loss: 0.006109
[INFO] Epoch 12/25 - Average loss: 0.005956
[INFO] Epoch 13/25 - Average loss: 0.005785
[INFO] Epoch 14/25 - Average loss: 0.005626
[INFO] Epoch 15/25 - Average loss: 0.005527
[INFO] Epoch 16/25 - Average loss: 0.005439
[INFO] Epoch 17/25 - Average loss: 0.005345
[INFO] Epoch 18/25 - Average loss: 0.005265
[INFO] Epoch 19/25 - Average loss: 0.005199
[INFO] Epoch 20/25 - Average loss: 0.005084
[INFO] Epoch 21/25 - Average loss: 0.005022
[INFO] Epoch 22/25 - Average loss: 0.005004
[INFO] Epoch 23/25 - Average loss: 0.004902
[INFO] Epoch 24/25 - Average loss: 0.004877
[INFO] Epoch 25/25 - Average loss: 0.004806
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.2332, 0.1435, 0.0387,
..., 0.0111, 0.0256, 0.0428])
[INFO] Number of well-explained examples: 2927/6386
[INFO] Iteration 2/3 for Translate
[INFO] Preparing DataLoader: batch_size=64, mask sum=2927
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.095856
[INFO] Epoch 2/25 - Average loss: 0.069806
[INFO] Epoch 3/25 - Average loss: 0.029066
[INFO] Epoch 4/25 - Average loss: 0.017890
[INFO] Epoch 5/25 - Average loss: 0.009899
[INFO] Epoch 6/25 - Average loss: 0.006938
[INFO] Epoch 7/25 - Average loss: 0.005796
[INFO] Epoch 8/25 - Average loss: 0.005330
[INFO] Epoch 9/25 - Average loss: 0.004960
[INFO] Epoch 10/25 - Average loss: 0.004684
[INFO] Epoch 11/25 - Average loss: 0.004389

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[INFO] Epoch 12/25 - Average loss: 0.004088
[INFO] Epoch 13/25 - Average loss: 0.003820
[INFO] Epoch 14/25 - Average loss: 0.003492
[INFO] Epoch 15/25 - Average loss: 0.003209
[INFO] Epoch 16/25 - Average loss: 0.002962
[INFO] Epoch 17/25 - Average loss: 0.002801
[INFO] Epoch 18/25 - Average loss: 0.002595
[INFO] Epoch 19/25 - Average loss: 0.002446
[INFO] Epoch 20/25 - Average loss: 0.002308
[INFO] Epoch 21/25 - Average loss: 0.002177
[INFO] Epoch 22/25 - Average loss: 0.002106
[INFO] Epoch 23/25 - Average loss: 0.002005
[INFO] Epoch 24/25 - Average loss: 0.001923
[INFO] Epoch 25/25 - Average loss: 0.001916
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.1065, 0.1606, 0.0541,
..., 0.0179, 0.0117, 0.0326])
[INFO] Number of well-explained examples: 2813/6386
[INFO] Iteration 3/3 for Translate
[INFO] Preparing DataLoader: batch_size=64, mask sum=2813
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.111194
[INFO] Epoch 2/25 - Average loss: 0.079265
[INFO] Epoch 3/25 - Average loss: 0.039364
[INFO] Epoch 4/25 - Average loss: 0.026694
[INFO] Epoch 5/25 - Average loss: 0.018825
[INFO] Epoch 6/25 - Average loss: 0.010655
[INFO] Epoch 7/25 - Average loss: 0.008813
[INFO] Epoch 8/25 - Average loss: 0.007047
[INFO] Epoch 9/25 - Average loss: 0.005454
[INFO] Epoch 10/25 - Average loss: 0.004220
[INFO] Epoch 11/25 - Average loss: 0.003274
[INFO] Epoch 12/25 - Average loss: 0.002560
[INFO] Epoch 13/25 - Average loss: 0.002064
[INFO] Epoch 14/25 - Average loss: 0.001632
[INFO] Epoch 15/25 - Average loss: 0.001302
[INFO] Epoch 16/25 - Average loss: 0.001113
[INFO] Epoch 17/25 - Average loss: 0.001010
[INFO] Epoch 18/25 - Average loss: 0.000872
[INFO] Epoch 19/25 - Average loss: 0.000807
[INFO] Epoch 20/25 - Average loss: 0.000755
[INFO] Epoch 21/25 - Average loss: 0.000690
[INFO] Epoch 22/25 - Average loss: 0.000649
[INFO] Epoch 23/25 - Average loss: 0.000605

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[INFO] Epoch 24/25 - Average loss: 0.000592
[INFO] Epoch 25/25 - Average loss: 0.000569
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0807, 0.2694, 0.0622,
..., 0.0549, 0.0117, 0.0159])
[INFO] Number of well-explained examples: 3157/6386
[SUCCESS] Model training finished for Translate. Total well-explained
examples: 3157/6386
[SUCCESS] Training model for SymRef with 1950 samples, 6 params
[INFO] Iteration 1/3 for SymRef
[INFO] Preparing DataLoader: batch_size=64, mask sum=1950
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.133051
[INFO] Epoch 2/25 - Average loss: 0.066060
[INFO] Epoch 3/25 - Average loss: 0.053947
[INFO] Epoch 4/25 - Average loss: 0.042203
[INFO] Epoch 5/25 - Average loss: 0.031015
[INFO] Epoch 6/25 - Average loss: 0.025494
[INFO] Epoch 7/25 - Average loss: 0.019167
[INFO] Epoch 8/25 - Average loss: 0.009996
[INFO] Epoch 9/25 - Average loss: 0.004409
[INFO] Epoch 10/25 - Average loss: 0.002262
[INFO] Epoch 11/25 - Average loss: 0.001350
[INFO] Epoch 12/25 - Average loss: 0.001167
[INFO] Epoch 13/25 - Average loss: 0.001103
[INFO] Epoch 14/25 - Average loss: 0.001065
[INFO] Epoch 15/25 - Average loss: 0.001030
[INFO] Epoch 16/25 - Average loss: 0.001000
[INFO] Epoch 17/25 - Average loss: 0.000966
[INFO] Epoch 18/25 - Average loss: 0.000957
[INFO] Epoch 19/25 - Average loss: 0.000944
[INFO] Epoch 20/25 - Average loss: 0.000909
[INFO] Epoch 21/25 - Average loss: 0.000900
[INFO] Epoch 22/25 - Average loss: 0.000877
[INFO] Epoch 23/25 - Average loss: 0.000880
[INFO] Epoch 24/25 - Average loss: 0.000869
[INFO] Epoch 25/25 - Average loss: 0.000859
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0200, 0.0078, 0.0075,
..., 0.0074, 0.0182, 0.0521])
[INFO] Number of well-explained examples: 1735/1950
[INFO] Iteration 2/3 for SymRef
[INFO] Preparing DataLoader: batch_size=64, mask sum=1735

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[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.166505
[INFO] Epoch 2/25 - Average loss: 0.071357
[INFO] Epoch 3/25 - Average loss: 0.039885
[INFO] Epoch 4/25 - Average loss: 0.038384
[INFO] Epoch 5/25 - Average loss: 0.035610
[INFO] Epoch 6/25 - Average loss: 0.027742
[INFO] Epoch 7/25 - Average loss: 0.019312
[INFO] Epoch 8/25 - Average loss: 0.011038
[INFO] Epoch 9/25 - Average loss: 0.003296
[INFO] Epoch 10/25 - Average loss: 0.002566
[INFO] Epoch 11/25 - Average loss: 0.002261
[INFO] Epoch 12/25 - Average loss: 0.001973
[INFO] Epoch 13/25 - Average loss: 0.001702
[INFO] Epoch 14/25 - Average loss: 0.001449
[INFO] Epoch 15/25 - Average loss: 0.001217
[INFO] Epoch 16/25 - Average loss: 0.001014
[INFO] Epoch 17/25 - Average loss: 0.000822
[INFO] Epoch 18/25 - Average loss: 0.000730
[INFO] Epoch 19/25 - Average loss: 0.000663
[INFO] Epoch 20/25 - Average loss: 0.000613
[INFO] Epoch 21/25 - Average loss: 0.000590
[INFO] Epoch 22/25 - Average loss: 0.000526
[INFO] Epoch 23/25 - Average loss: 0.000496
[INFO] Epoch 24/25 - Average loss: 0.000468
[INFO] Epoch 25/25 - Average loss: 0.000444
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0161, 0.0075, 0.0049,
..., 0.0058, 0.0137, 0.0505])
[INFO] Number of well-explained examples: 1722/1950
[INFO] Iteration 3/3 for SymRef
[INFO] Preparing DataLoader: batch_size=64, mask sum=1722
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.178328
[INFO] Epoch 2/25 - Average loss: 0.100711
[INFO] Epoch 3/25 - Average loss: 0.050356
[INFO] Epoch 4/25 - Average loss: 0.036352
[INFO] Epoch 5/25 - Average loss: 0.034886
[INFO] Epoch 6/25 - Average loss: 0.031239
[INFO] Epoch 7/25 - Average loss: 0.020390
[INFO] Epoch 8/25 - Average loss: 0.013511

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[INFO] Epoch 9/25 - Average loss: 0.012288
[INFO] Epoch 10/25 - Average loss: 0.011465
[INFO] Epoch 11/25 - Average loss: 0.010561
[INFO] Epoch 12/25 - Average loss: 0.009714
[INFO] Epoch 13/25 - Average loss: 0.007945
[INFO] Epoch 14/25 - Average loss: 0.004002
[INFO] Epoch 15/25 - Average loss: 0.001182
[INFO] Epoch 16/25 - Average loss: 0.000431
[INFO] Epoch 17/25 - Average loss: 0.000264
[INFO] Epoch 18/25 - Average loss: 0.000189
[INFO] Epoch 19/25 - Average loss: 0.000148
[INFO] Epoch 20/25 - Average loss: 0.000124
[INFO] Epoch 21/25 - Average loss: 0.000107
[INFO] Epoch 22/25 - Average loss: 0.000095
[INFO] Epoch 23/25 - Average loss: 0.000087
[INFO] Epoch 24/25 - Average loss: 0.000080
[INFO] Epoch 25/25 - Average loss: 0.000075
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0171, 0.0053, 0.0067,
..., 0.0066, 0.0138, 0.0489])
[INFO] Number of well-explained examples: 1715/1950
[SUCCESS] Model training finished for SymRef. Total well-explained
examples: 1715/1950
[INFO] Skipping SymTrans (only 42 examples)
[INFO] Skipping SymRot (only 23 examples)
[SUCCESS] Finished training 4 SINGLETONS
[INFO] Starting pair model training...
[INFO] Finding abstractions for 12 structures of type PAIRS
[SUCCESS] Training model for Rotate(Scale) with 6386 samples, 7 params
[INFO] Iteration 1/3 for Rotate(Scale)
[INFO] Preparing DataLoader: batch_size=64, mask sum=6386
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=7, hidden_dim=6
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.128704
[INFO] Epoch 2/25 - Average loss: 0.032779
[INFO] Epoch 3/25 - Average loss: 0.012408
[INFO] Epoch 4/25 - Average loss: 0.003430
[INFO] Epoch 5/25 - Average loss: 0.001901
[INFO] Epoch 6/25 - Average loss: 0.001886
[INFO] Epoch 7/25 - Average loss: 0.001862
[INFO] Epoch 8/25 - Average loss: 0.001866
[INFO] Epoch 9/25 - Average loss: 0.001854
[INFO] Epoch 10/25 - Average loss: 0.001854
[INFO] Epoch 11/25 - Average loss: 0.001836

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[INFO] Epoch 12/25 - Average loss: 0.001828
[INFO] Epoch 13/25 - Average loss: 0.001826
[INFO] Epoch 14/25 - Average loss: 0.001797
[INFO] Epoch 15/25 - Average loss: 0.001771
[INFO] Epoch 16/25 - Average loss: 0.001730
[INFO] Epoch 17/25 - Average loss: 0.001615
[INFO] Epoch 18/25 - Average loss: 0.001426
[INFO] Epoch 19/25 - Average loss: 0.001238
[INFO] Epoch 20/25 - Average loss: 0.001182
[INFO] Epoch 21/25 - Average loss: 0.001160
[INFO] Epoch 22/25 - Average loss: 0.001139
[INFO] Epoch 23/25 - Average loss: 0.001130
[INFO] Epoch 24/25 - Average loss: 0.001115
[INFO] Epoch 25/25 - Average loss: 0.001099
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0169, 0.3304, 0.0645,
..., 0.0038, 0.0042, 0.0501])
[INFO] Number of well-explained examples: 4659/6386
[INFO] Iteration 2/3 for Rotate(Scale)
[INFO] Preparing DataLoader: batch_size=64, mask sum=4659
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=7, hidden_dim=6
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.163630
[INFO] Epoch 2/25 - Average loss: 0.045387
[INFO] Epoch 3/25 - Average loss: 0.031895
[INFO] Epoch 4/25 - Average loss: 0.022705
[INFO] Epoch 5/25 - Average loss: 0.014345
[INFO] Epoch 6/25 - Average loss: 0.010709
[INFO] Epoch 7/25 - Average loss: 0.007617
[INFO] Epoch 8/25 - Average loss: 0.001158
[INFO] Epoch 9/25 - Average loss: 0.000386
[INFO] Epoch 10/25 - Average loss: 0.000365
[INFO] Epoch 11/25 - Average loss: 0.000360
[INFO] Epoch 12/25 - Average loss: 0.000346
[INFO] Epoch 13/25 - Average loss: 0.000342
[INFO] Epoch 14/25 - Average loss: 0.000325
[INFO] Epoch 15/25 - Average loss: 0.000283
[INFO] Epoch 16/25 - Average loss: 0.000224
[INFO] Epoch 17/25 - Average loss: 0.000169
[INFO] Epoch 18/25 - Average loss: 0.000102
[INFO] Epoch 19/25 - Average loss: 0.000084
[INFO] Epoch 20/25 - Average loss: 0.000078
[INFO] Epoch 21/25 - Average loss: 0.000073
[INFO] Epoch 22/25 - Average loss: 0.000072
[INFO] Epoch 23/25 - Average loss: 0.000072

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[INFO] Epoch 24/25 - Average loss: 0.000072
[INFO] Epoch 25/25 - Average loss: 0.000071
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0035, 0.3375, 0.0647,
..., 0.0029, 0.0035, 0.0120])
[INFO] Number of well-explained examples: 4711/6386
[INFO] Iteration 3/3 for Rotate(Scale)
[INFO] Preparing DataLoader: batch_size=64, mask sum=4711
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=7, hidden_dim=6
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.140131
[INFO] Epoch 2/25 - Average loss: 0.045923
[INFO] Epoch 3/25 - Average loss: 0.029521
[INFO] Epoch 4/25 - Average loss: 0.024626
[INFO] Epoch 5/25 - Average loss: 0.017899
[INFO] Epoch 6/25 - Average loss: 0.009116
[INFO] Epoch 7/25 - Average loss: 0.005690
[INFO] Epoch 8/25 - Average loss: 0.000644
[INFO] Epoch 9/25 - Average loss: 0.000315
[INFO] Epoch 10/25 - Average loss: 0.000305
[INFO] Epoch 11/25 - Average loss: 0.000296
[INFO] Epoch 12/25 - Average loss: 0.000295
[INFO] Epoch 13/25 - Average loss: 0.000290
[INFO] Epoch 14/25 - Average loss: 0.000295
[INFO] Epoch 15/25 - Average loss: 0.000294
[INFO] Epoch 16/25 - Average loss: 0.000290
[INFO] Epoch 17/25 - Average loss: 0.000290
[INFO] Epoch 18/25 - Average loss: 0.000291
[INFO] Epoch 19/25 - Average loss: 0.000291
[INFO] Epoch 20/25 - Average loss: 0.000288
[INFO] Epoch 21/25 - Average loss: 0.000292
[INFO] Epoch 22/25 - Average loss: 0.000290
[INFO] Epoch 23/25 - Average loss: 0.000291
[INFO] Epoch 24/25 - Average loss: 0.000295
[INFO] Epoch 25/25 - Average loss: 0.000287
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0043, 0.3195, 0.0718,
..., 0.0016, 0.0018, 0.0304])
[INFO] Number of well-explained examples: 4414/6386
[SUCCESS] Model training finished for Rotate(Scale). Total well-explained
examples: 4414/6386
[SUCCESS] Training model for Scale(Box) with 6386 samples, 3 params
[INFO] Iteration 1/3 for Scale(Box)
[INFO] Preparing DataLoader: batch_size=64, mask sum=6386

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[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.136426
[INFO] Epoch 2/25 - Average loss: 0.084568
[INFO] Epoch 3/25 - Average loss: 0.041243
[INFO] Epoch 4/25 - Average loss: 0.024901
[INFO] Epoch 5/25 - Average loss: 0.015534
[INFO] Epoch 6/25 - Average loss: 0.010212
[INFO] Epoch 7/25 - Average loss: 0.008716
[INFO] Epoch 8/25 - Average loss: 0.007832
[INFO] Epoch 9/25 - Average loss: 0.007241
[INFO] Epoch 10/25 - Average loss: 0.006861
[INFO] Epoch 11/25 - Average loss: 0.006217
[INFO] Epoch 12/25 - Average loss: 0.005813
[INFO] Epoch 13/25 - Average loss: 0.005677
[INFO] Epoch 14/25 - Average loss: 0.005290
[INFO] Epoch 15/25 - Average loss: 0.005070
[INFO] Epoch 16/25 - Average loss: 0.005016
[INFO] Epoch 17/25 - Average loss: 0.004887
[INFO] Epoch 18/25 - Average loss: 0.004744
[INFO] Epoch 19/25 - Average loss: 0.004694
[INFO] Epoch 20/25 - Average loss: 0.004632
[INFO] Epoch 21/25 - Average loss: 0.004584
[INFO] Epoch 22/25 - Average loss: 0.004458
[INFO] Epoch 23/25 - Average loss: 0.004420
[INFO] Epoch 24/25 - Average loss: 0.004341
[INFO] Epoch 25/25 - Average loss: 0.004320
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0230, 0.1775, 0.0654,
..., 0.0086, 0.0035, 0.0139])
[INFO] Number of well-explained examples: 4111/6386
[INFO] Iteration 2/3 for Scale(Box)
[INFO] Preparing DataLoader: batch_size=64, mask sum=4111
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.112189
[INFO] Epoch 2/25 - Average loss: 0.061875
[INFO] Epoch 3/25 - Average loss: 0.038553
[INFO] Epoch 4/25 - Average loss: 0.022717
[INFO] Epoch 5/25 - Average loss: 0.017105
[INFO] Epoch 6/25 - Average loss: 0.013943
[INFO] Epoch 7/25 - Average loss: 0.011191
[INFO] Epoch 8/25 - Average loss: 0.008967

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[INFO] Epoch 9/25 - Average loss: 0.007647
[INFO] Epoch 10/25 - Average loss: 0.006502
[INFO] Epoch 11/25 - Average loss: 0.005677
[INFO] Epoch 12/25 - Average loss: 0.004770
[INFO] Epoch 13/25 - Average loss: 0.004221
[INFO] Epoch 14/25 - Average loss: 0.003835
[INFO] Epoch 15/25 - Average loss: 0.003509
[INFO] Epoch 16/25 - Average loss: 0.003297
[INFO] Epoch 17/25 - Average loss: 0.003070
[INFO] Epoch 18/25 - Average loss: 0.002856
[INFO] Epoch 19/25 - Average loss: 0.002828
[INFO] Epoch 20/25 - Average loss: 0.002610
[INFO] Epoch 21/25 - Average loss: 0.002454
[INFO] Epoch 22/25 - Average loss: 0.002382
[INFO] Epoch 23/25 - Average loss: 0.002229
[INFO] Epoch 24/25 - Average loss: 0.002153
[INFO] Epoch 25/25 - Average loss: 0.002037
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0238, 0.0683, 0.1068,
..., 0.0042, 0.0158, 0.0195])
[INFO] Number of well-explained examples: 3367/6386
[INFO] Iteration 3/3 for Scale(Box)
[INFO] Preparing DataLoader: batch_size=64, mask sum=3367
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.113164
[INFO] Epoch 2/25 - Average loss: 0.048373
[INFO] Epoch 3/25 - Average loss: 0.021798
[INFO] Epoch 4/25 - Average loss: 0.009056
[INFO] Epoch 5/25 - Average loss: 0.006898
[INFO] Epoch 6/25 - Average loss: 0.004753
[INFO] Epoch 7/25 - Average loss: 0.002632
[INFO] Epoch 8/25 - Average loss: 0.001562
[INFO] Epoch 9/25 - Average loss: 0.001221
[INFO] Epoch 10/25 - Average loss: 0.001049
[INFO] Epoch 11/25 - Average loss: 0.000947
[INFO] Epoch 12/25 - Average loss: 0.000827
[INFO] Epoch 13/25 - Average loss: 0.000764
[INFO] Epoch 14/25 - Average loss: 0.000694
[INFO] Epoch 15/25 - Average loss: 0.000699
[INFO] Epoch 16/25 - Average loss: 0.000646
[INFO] Epoch 17/25 - Average loss: 0.000611
[INFO] Epoch 18/25 - Average loss: 0.000574
[INFO] Epoch 19/25 - Average loss: 0.000562
[INFO] Epoch 20/25 - Average loss: 0.000550

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[INFO] Epoch 21/25 - Average loss: 0.000532
[INFO] Epoch 22/25 - Average loss: 0.000542
[INFO] Epoch 23/25 - Average loss: 0.000541
[INFO] Epoch 24/25 - Average loss: 0.000521
[INFO] Epoch 25/25 - Average loss: 0.000506
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0312, 0.1633, 0.1218,
..., 0.0137, 0.0169, 0.0520])
[INFO] Number of well-explained examples: 3829/6386
[SUCCESS] Model training finished for Scale(Box). Total well-explained
examples: 3829/6386
[SUCCESS] Training model for Translate(Rotate) with 6386 samples, 7
params
[INFO] Iteration 1/3 for Translate(Rotate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=6386
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=7, hidden_dim=6
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.092918
[INFO] Epoch 2/25 - Average loss: 0.023385
[INFO] Epoch 3/25 - Average loss: 0.015043
[INFO] Epoch 4/25 - Average loss: 0.010407
[INFO] Epoch 5/25 - Average loss: 0.005993
[INFO] Epoch 6/25 - Average loss: 0.002044
[INFO] Epoch 7/25 - Average loss: 0.001927
[INFO] Epoch 8/25 - Average loss: 0.001902
[INFO] Epoch 9/25 - Average loss: 0.001922
[INFO] Epoch 10/25 - Average loss: 0.001877
[INFO] Epoch 11/25 - Average loss: 0.001834
[INFO] Epoch 12/25 - Average loss: 0.001826
[INFO] Epoch 13/25 - Average loss: 0.001796
[INFO] Epoch 14/25 - Average loss: 0.001752
[INFO] Epoch 15/25 - Average loss: 0.001712
[INFO] Epoch 16/25 - Average loss: 0.001604
[INFO] Epoch 17/25 - Average loss: 0.001461
[INFO] Epoch 18/25 - Average loss: 0.001272
[INFO] Epoch 19/25 - Average loss: 0.001115
[INFO] Epoch 20/25 - Average loss: 0.001040
[INFO] Epoch 21/25 - Average loss: 0.001007
[INFO] Epoch 22/25 - Average loss: 0.000944
[INFO] Epoch 23/25 - Average loss: 0.000728
[INFO] Epoch 24/25 - Average loss: 0.000364
[INFO] Epoch 25/25 - Average loss: 0.000309
[SUCCESS] Autoencoder training complete.

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[INFO] Max reconstruction error per sample: tensor([0.0133, 0.0342, 0.0440,
..., 0.0085, 0.0165, 0.0150])
[INFO] Number of well-explained examples: 5832/6386
[INFO] Iteration 2/3 for Translate(Rotate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=5832
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=7, hidden_dim=6
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.084329
[INFO] Epoch 2/25 - Average loss: 0.026663
[INFO] Epoch 3/25 - Average loss: 0.012642
[INFO] Epoch 4/25 - Average loss: 0.008880
[INFO] Epoch 5/25 - Average loss: 0.003331
[INFO] Epoch 6/25 - Average loss: 0.001091
[INFO] Epoch 7/25 - Average loss: 0.001035
[INFO] Epoch 8/25 - Average loss: 0.000995
[INFO] Epoch 9/25 - Average loss: 0.000951
[INFO] Epoch 10/25 - Average loss: 0.000835
[INFO] Epoch 11/25 - Average loss: 0.000602
[INFO] Epoch 12/25 - Average loss: 0.000411
[INFO] Epoch 13/25 - Average loss: 0.000372
[INFO] Epoch 14/25 - Average loss: 0.000349
[INFO] Epoch 15/25 - Average loss: 0.000342
[INFO] Epoch 16/25 - Average loss: 0.000342
[INFO] Epoch 17/25 - Average loss: 0.000338
[INFO] Epoch 18/25 - Average loss: 0.000342
[INFO] Epoch 19/25 - Average loss: 0.000334
[INFO] Epoch 20/25 - Average loss: 0.000332
[INFO] Epoch 21/25 - Average loss: 0.000332
[INFO] Epoch 22/25 - Average loss: 0.000335
[INFO] Epoch 23/25 - Average loss: 0.000336
[INFO] Epoch 24/25 - Average loss: 0.000333
[INFO] Epoch 25/25 - Average loss: 0.000330
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0088, 0.0298, 0.0472,
..., 0.0121, 0.0179, 0.0342])
[INFO] Number of well-explained examples: 5341/6386
[INFO] Iteration 3/3 for Translate(Rotate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=5341
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=7, hidden_dim=6
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.096939
[INFO] Epoch 2/25 - Average loss: 0.021250

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[INFO] Epoch 3/25 - Average loss: 0.014263
[INFO] Epoch 4/25 - Average loss: 0.006864
[INFO] Epoch 5/25 - Average loss: 0.001599
[INFO] Epoch 6/25 - Average loss: 0.000713
[INFO] Epoch 7/25 - Average loss: 0.000655
[INFO] Epoch 8/25 - Average loss: 0.000624
[INFO] Epoch 9/25 - Average loss: 0.000608
[INFO] Epoch 10/25 - Average loss: 0.000599
[INFO] Epoch 11/25 - Average loss: 0.000584
[INFO] Epoch 12/25 - Average loss: 0.000591
[INFO] Epoch 13/25 - Average loss: 0.000559
[INFO] Epoch 14/25 - Average loss: 0.000534
[INFO] Epoch 15/25 - Average loss: 0.000485
[INFO] Epoch 16/25 - Average loss: 0.000414
[INFO] Epoch 17/25 - Average loss: 0.000280
[INFO] Epoch 18/25 - Average loss: 0.000125
[INFO] Epoch 19/25 - Average loss: 0.000048
[INFO] Epoch 20/25 - Average loss: 0.000035
[INFO] Epoch 21/25 - Average loss: 0.000031
[INFO] Epoch 22/25 - Average loss: 0.000029
[INFO] Epoch 23/25 - Average loss: 0.000027
[INFO] Epoch 24/25 - Average loss: 0.000027
[INFO] Epoch 25/25 - Average loss: 0.000028
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0041, 0.0282, 0.0529,
..., 0.0042, 0.0036, 0.0236])
[INFO] Number of well-explained examples: 5360/6386
[SUCCESS] Model training finished for Translate(Rotate). Total well-
explained examples: 5360/6386
[SUCCESS] Training model for Union(Translate) with 5513 samples, 3
params
[INFO] Iteration 1/3 for Union(Translate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=5513
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.089776
[INFO] Epoch 2/25 - Average loss: 0.040754
[INFO] Epoch 3/25 - Average loss: 0.034792
[INFO] Epoch 4/25 - Average loss: 0.019696
[INFO] Epoch 5/25 - Average loss: 0.015854
[INFO] Epoch 6/25 - Average loss: 0.014559
[INFO] Epoch 7/25 - Average loss: 0.013564
[INFO] Epoch 8/25 - Average loss: 0.012912
[INFO] Epoch 9/25 - Average loss: 0.012333

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[INFO] Epoch 10/25 - Average loss: 0.011770
[INFO] Epoch 11/25 - Average loss: 0.011204
[INFO] Epoch 12/25 - Average loss: 0.010562
[INFO] Epoch 13/25 - Average loss: 0.009941
[INFO] Epoch 14/25 - Average loss: 0.009196
[INFO] Epoch 15/25 - Average loss: 0.008562
[INFO] Epoch 16/25 - Average loss: 0.007957
[INFO] Epoch 17/25 - Average loss: 0.007482
[INFO] Epoch 18/25 - Average loss: 0.007054
[INFO] Epoch 19/25 - Average loss: 0.006835
[INFO] Epoch 20/25 - Average loss: 0.006630
[INFO] Epoch 21/25 - Average loss: 0.006435
[INFO] Epoch 22/25 - Average loss: 0.006270
[INFO] Epoch 23/25 - Average loss: 0.006162
[INFO] Epoch 24/25 - Average loss: 0.006141
[INFO] Epoch 25/25 - Average loss: 0.006017
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.1269, 0.1063, 0.0494,
..., 0.0107, 0.0175, 0.0048])
[INFO] Number of well-explained examples: 2819/5513
[INFO] Iteration 2/3 for Union(Translate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=2819
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.105195
[INFO] Epoch 2/25 - Average loss: 0.068682
[INFO] Epoch 3/25 - Average loss: 0.031910
[INFO] Epoch 4/25 - Average loss: 0.023663
[INFO] Epoch 5/25 - Average loss: 0.020870
[INFO] Epoch 6/25 - Average loss: 0.014491
[INFO] Epoch 7/25 - Average loss: 0.007301
[INFO] Epoch 8/25 - Average loss: 0.005934
[INFO] Epoch 9/25 - Average loss: 0.005524
[INFO] Epoch 10/25 - Average loss: 0.005051
[INFO] Epoch 11/25 - Average loss: 0.004784
[INFO] Epoch 12/25 - Average loss: 0.004629
[INFO] Epoch 13/25 - Average loss: 0.004359
[INFO] Epoch 14/25 - Average loss: 0.004099
[INFO] Epoch 15/25 - Average loss: 0.004019
[INFO] Epoch 16/25 - Average loss: 0.003765
[INFO] Epoch 17/25 - Average loss: 0.003824
[INFO] Epoch 18/25 - Average loss: 0.003379
[INFO] Epoch 19/25 - Average loss: 0.003266
[INFO] Epoch 20/25 - Average loss: 0.003164
[INFO] Epoch 21/25 - Average loss: 0.003091

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[INFO] Epoch 22/25 - Average loss: 0.003006
[INFO] Epoch 23/25 - Average loss: 0.002860
[INFO] Epoch 24/25 - Average loss: 0.002830
[INFO] Epoch 25/25 - Average loss: 0.002726
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0578, 0.1770, 0.0434,
..., 0.0210, 0.0039, 0.0159])
[INFO] Number of well-explained examples: 2301/5513
[INFO] Iteration 3/3 for Union(Translate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=2301
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=3, hidden_dim=2
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.079457
[INFO] Epoch 2/25 - Average loss: 0.062093
[INFO] Epoch 3/25 - Average loss: 0.027689
[INFO] Epoch 4/25 - Average loss: 0.017713
[INFO] Epoch 5/25 - Average loss: 0.015453
[INFO] Epoch 6/25 - Average loss: 0.013259
[INFO] Epoch 7/25 - Average loss: 0.010628
[INFO] Epoch 8/25 - Average loss: 0.008080
[INFO] Epoch 9/25 - Average loss: 0.006464
[INFO] Epoch 10/25 - Average loss: 0.005500
[INFO] Epoch 11/25 - Average loss: 0.005066
[INFO] Epoch 12/25 - Average loss: 0.004727
[INFO] Epoch 13/25 - Average loss: 0.004433
[INFO] Epoch 14/25 - Average loss: 0.004123
[INFO] Epoch 15/25 - Average loss: 0.003819
[INFO] Epoch 16/25 - Average loss: 0.003511
[INFO] Epoch 17/25 - Average loss: 0.003177
[INFO] Epoch 18/25 - Average loss: 0.002911
[INFO] Epoch 19/25 - Average loss: 0.002690
[INFO] Epoch 20/25 - Average loss: 0.002494
[INFO] Epoch 21/25 - Average loss: 0.002345
[INFO] Epoch 22/25 - Average loss: 0.002207
[INFO] Epoch 23/25 - Average loss: 0.002097
[INFO] Epoch 24/25 - Average loss: 0.002014
[INFO] Epoch 25/25 - Average loss: 0.001933
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0971, 0.1429, 0.0326,
..., 0.0114, 0.0218, 0.0401])
[INFO] Number of well-explained examples: 1737/5513
[SUCCESS] Model training finished for Union(Translate). Total well-
explained examples: 1737/5513
[SUCCESS] Training model for Union(SymRef) with 1950 samples, 6 params

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[INFO] Iteration 1/3 for Union(SymRef)
[INFO] Preparing DataLoader: batch_size=64, mask sum=1950
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.193585
[INFO] Epoch 2/25 - Average loss: 0.108359
[INFO] Epoch 3/25 - Average loss: 0.053694
[INFO] Epoch 4/25 - Average loss: 0.037734
[INFO] Epoch 5/25 - Average loss: 0.027245
[INFO] Epoch 6/25 - Average loss: 0.018604
[INFO] Epoch 7/25 - Average loss: 0.010262
[INFO] Epoch 8/25 - Average loss: 0.003495
[INFO] Epoch 9/25 - Average loss: 0.001470
[INFO] Epoch 10/25 - Average loss: 0.001250
[INFO] Epoch 11/25 - Average loss: 0.001194
[INFO] Epoch 12/25 - Average loss: 0.001152
[INFO] Epoch 13/25 - Average loss: 0.001127
[INFO] Epoch 14/25 - Average loss: 0.001096
[INFO] Epoch 15/25 - Average loss: 0.001063
[INFO] Epoch 16/25 - Average loss: 0.001044
[INFO] Epoch 17/25 - Average loss: 0.001017
[INFO] Epoch 18/25 - Average loss: 0.000981
[INFO] Epoch 19/25 - Average loss: 0.000975
[INFO] Epoch 20/25 - Average loss: 0.000955
[INFO] Epoch 21/25 - Average loss: 0.000926
[INFO] Epoch 22/25 - Average loss: 0.000905
[INFO] Epoch 23/25 - Average loss: 0.000881
[INFO] Epoch 24/25 - Average loss: 0.000869
[INFO] Epoch 25/25 - Average loss: 0.000845
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0223, 0.0070, 0.0108,
..., 0.0124, 0.0151, 0.0412])
[INFO] Number of well-explained examples: 1748/1950
[INFO] Iteration 2/3 for Union(SymRef)
[INFO] Preparing DataLoader: batch_size=64, mask sum=1748
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.148362
[INFO] Epoch 2/25 - Average loss: 0.078113
[INFO] Epoch 3/25 - Average loss: 0.044236
[INFO] Epoch 4/25 - Average loss: 0.041111
[INFO] Epoch 5/25 - Average loss: 0.040221
[INFO] Epoch 6/25 - Average loss: 0.037840

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[INFO] Epoch 7/25 - Average loss: 0.030346
[INFO] Epoch 8/25 - Average loss: 0.018487
[INFO] Epoch 9/25 - Average loss: 0.015440
[INFO] Epoch 10/25 - Average loss: 0.014379
[INFO] Epoch 11/25 - Average loss: 0.013062
[INFO] Epoch 12/25 - Average loss: 0.010446
[INFO] Epoch 13/25 - Average loss: 0.005820
[INFO] Epoch 14/25 - Average loss: 0.001725
[INFO] Epoch 15/25 - Average loss: 0.000539
[INFO] Epoch 16/25 - Average loss: 0.000358
[INFO] Epoch 17/25 - Average loss: 0.000292
[INFO] Epoch 18/25 - Average loss: 0.000258
[INFO] Epoch 19/25 - Average loss: 0.000240
[INFO] Epoch 20/25 - Average loss: 0.000201
[INFO] Epoch 21/25 - Average loss: 0.000173
[INFO] Epoch 22/25 - Average loss: 0.000149
[INFO] Epoch 23/25 - Average loss: 0.000130
[INFO] Epoch 24/25 - Average loss: 0.000113
[INFO] Epoch 25/25 - Average loss: 0.000103
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0149, 0.0055, 0.0054,
..., 0.0052, 0.0150, 0.0515])
[INFO] Number of well-explained examples: 1756/1950
[INFO] Iteration 3/3 for Union(SymRef)
[INFO] Preparing DataLoader: batch_size=64, mask sum=1756
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.154383
[INFO] Epoch 2/25 - Average loss: 0.073502
[INFO] Epoch 3/25 - Average loss: 0.042254
[INFO] Epoch 4/25 - Average loss: 0.038255
[INFO] Epoch 5/25 - Average loss: 0.032475
[INFO] Epoch 6/25 - Average loss: 0.024468
[INFO] Epoch 7/25 - Average loss: 0.015471
[INFO] Epoch 8/25 - Average loss: 0.008229
[INFO] Epoch 9/25 - Average loss: 0.004218
[INFO] Epoch 10/25 - Average loss: 0.003141
[INFO] Epoch 11/25 - Average loss: 0.002794
[INFO] Epoch 12/25 - Average loss: 0.002367
[INFO] Epoch 13/25 - Average loss: 0.001717
[INFO] Epoch 14/25 - Average loss: 0.001026
[INFO] Epoch 15/25 - Average loss: 0.000484
[INFO] Epoch 16/25 - Average loss: 0.000241
[INFO] Epoch 17/25 - Average loss: 0.000137
[INFO] Epoch 18/25 - Average loss: 0.000100

```

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[INFO] Epoch 19/25 - Average loss: 0.000088
[INFO] Epoch 20/25 - Average loss: 0.000082
[INFO] Epoch 21/25 - Average loss: 0.000080
[INFO] Epoch 22/25 - Average loss: 0.000075
[INFO] Epoch 23/25 - Average loss: 0.000075
[INFO] Epoch 24/25 - Average loss: 0.000073
[INFO] Epoch 25/25 - Average loss: 0.000072
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0177, 0.0068, 0.0053,
..., 0.0068, 0.0095, 0.0461])
[INFO] Number of well-explained examples: 1763/1950
[SUCCESS] Model training finished for Union(SymRef). Total well-explained
examples: 1763/1950
[SUCCESS] Training model for SymRef(Union) with 1128 samples, 6 params
[INFO] Iteration 1/3 for SymRef(Union)
[INFO] Preparing DataLoader: batch_size=64, mask sum=1128
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.236016
[INFO] Epoch 2/25 - Average loss: 0.173954
[INFO] Epoch 3/25 - Average loss: 0.123665
[INFO] Epoch 4/25 - Average loss: 0.083329
[INFO] Epoch 5/25 - Average loss: 0.051487
[INFO] Epoch 6/25 - Average loss: 0.044915
[INFO] Epoch 7/25 - Average loss: 0.042768
[INFO] Epoch 8/25 - Average loss: 0.040334
[INFO] Epoch 9/25 - Average loss: 0.035762
[INFO] Epoch 10/25 - Average loss: 0.028699
[INFO] Epoch 11/25 - Average loss: 0.021425
[INFO] Epoch 12/25 - Average loss: 0.014986
[INFO] Epoch 13/25 - Average loss: 0.007351
[INFO] Epoch 14/25 - Average loss: 0.004246
[INFO] Epoch 15/25 - Average loss: 0.003139
[INFO] Epoch 16/25 - Average loss: 0.002218
[INFO] Epoch 17/25 - Average loss: 0.001542
[INFO] Epoch 18/25 - Average loss: 0.000984
[INFO] Epoch 19/25 - Average loss: 0.000745
[INFO] Epoch 20/25 - Average loss: 0.000611
[INFO] Epoch 21/25 - Average loss: 0.000547
[INFO] Epoch 22/25 - Average loss: 0.000515
[INFO] Epoch 23/25 - Average loss: 0.000489
[INFO] Epoch 24/25 - Average loss: 0.000467
[INFO] Epoch 25/25 - Average loss: 0.000456
[SUCCESS] Autoencoder training complete.

```

```

[INFO] Max reconstruction error per sample: tensor([0.0140, 0.0069, 0.0185,
..., 0.0069, 0.0172, 0.0497])
[INFO] Number of well-explained examples: 1040/1128
[INFO] Iteration 2/3 for SymRef(Union)
[INFO] Preparing DataLoader: batch_size=64, mask sum=1040
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.152044
[INFO] Epoch 2/25 - Average loss: 0.111142
[INFO] Epoch 3/25 - Average loss: 0.074781
[INFO] Epoch 4/25 - Average loss: 0.045058
[INFO] Epoch 5/25 - Average loss: 0.036728
[INFO] Epoch 6/25 - Average loss: 0.036206
[INFO] Epoch 7/25 - Average loss: 0.035709
[INFO] Epoch 8/25 - Average loss: 0.034701
[INFO] Epoch 9/25 - Average loss: 0.032544
[INFO] Epoch 10/25 - Average loss: 0.027495
[INFO] Epoch 11/25 - Average loss: 0.021684
[INFO] Epoch 12/25 - Average loss: 0.019102
[INFO] Epoch 13/25 - Average loss: 0.018487
[INFO] Epoch 14/25 - Average loss: 0.017912
[INFO] Epoch 15/25 - Average loss: 0.017406
[INFO] Epoch 16/25 - Average loss: 0.016972
[INFO] Epoch 17/25 - Average loss: 0.016657
[INFO] Epoch 18/25 - Average loss: 0.016214
[INFO] Epoch 19/25 - Average loss: 0.015456
[INFO] Epoch 20/25 - Average loss: 0.013428
[INFO] Epoch 21/25 - Average loss: 0.007605
[INFO] Epoch 22/25 - Average loss: 0.002206
[INFO] Epoch 23/25 - Average loss: 0.000982
[INFO] Epoch 24/25 - Average loss: 0.000657
[INFO] Epoch 25/25 - Average loss: 0.000509
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0190, 0.0064, 0.0140,
..., 0.0064, 0.0108, 0.0429])
[INFO] Number of well-explained examples: 951/1128
[INFO] Iteration 3/3 for SymRef(Union)
[INFO] Preparing DataLoader: batch_size=64, mask sum=951
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=6, hidden_dim=5
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.266325
[INFO] Epoch 2/25 - Average loss: 0.216758

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```

[INFO] Epoch 3/25 - Average loss: 0.159518
[INFO] Epoch 4/25 - Average loss: 0.092815
[INFO] Epoch 5/25 - Average loss: 0.040339
[INFO] Epoch 6/25 - Average loss: 0.034657
[INFO] Epoch 7/25 - Average loss: 0.032796
[INFO] Epoch 8/25 - Average loss: 0.032353
[INFO] Epoch 9/25 - Average loss: 0.031920
[INFO] Epoch 10/25 - Average loss: 0.031288
[INFO] Epoch 11/25 - Average loss: 0.030362
[INFO] Epoch 12/25 - Average loss: 0.028819
[INFO] Epoch 13/25 - Average loss: 0.026238
[INFO] Epoch 14/25 - Average loss: 0.022596
[INFO] Epoch 15/25 - Average loss: 0.018660
[INFO] Epoch 16/25 - Average loss: 0.015988
[INFO] Epoch 17/25 - Average loss: 0.014714
[INFO] Epoch 18/25 - Average loss: 0.013630
[INFO] Epoch 19/25 - Average loss: 0.012349
[INFO] Epoch 20/25 - Average loss: 0.010522
[INFO] Epoch 21/25 - Average loss: 0.008073
[INFO] Epoch 22/25 - Average loss: 0.004755
[INFO] Epoch 23/25 - Average loss: 0.001379
[INFO] Epoch 24/25 - Average loss: 0.000229
[INFO] Epoch 25/25 - Average loss: 0.000158
[SUCCESS] Autoencoder training complete.
[INFO] Max reconstruction error per sample: tensor([0.0204, 0.0098, 0.0122,
..., 0.0098, 0.0171, 0.0502])
[INFO] Number of well-explained examples: 965/1128
[SUCCESS] Model training finished for SymRef(Union). Total well-explained
examples: 965/1128
[SUCCESS] Training model for SymRef(Translate) with 822 samples, 9
params
[INFO] Iteration 1/3 for SymRef(Translate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=822
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=9, hidden_dim=8
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.180439
[INFO] Epoch 2/25 - Average loss: 0.157405
[INFO] Epoch 3/25 - Average loss: 0.141729
[INFO] Epoch 4/25 - Average loss: 0.126745
[INFO] Epoch 5/25 - Average loss: 0.108196
[INFO] Epoch 6/25 - Average loss: 0.082494
[INFO] Epoch 7/25 - Average loss: 0.066343
[INFO] Epoch 8/25 - Average loss: 0.058807
[INFO] Epoch 9/25 - Average loss: 0.050892

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```
[INFO] Epoch 10/25 - Average loss: 0.043409
[INFO] Epoch 11/25 - Average loss: 0.037483
[INFO] Epoch 12/25 - Average loss: 0.033402
[INFO] Epoch 13/25 - Average loss: 0.029725
[INFO] Epoch 14/25 - Average loss: 0.025876
[INFO] Epoch 15/25 - Average loss: 0.021670
[INFO] Epoch 16/25 - Average loss: 0.017294
[INFO] Epoch 17/25 - Average loss: 0.013286
[INFO] Epoch 18/25 - Average loss: 0.010369
[INFO] Epoch 19/25 - Average loss: 0.008675
[INFO] Epoch 20/25 - Average loss: 0.007711
[INFO] Epoch 21/25 - Average loss: 0.007162
[INFO] Epoch 22/25 - Average loss: 0.006834
[INFO] Epoch 23/25 - Average loss: 0.006531
[INFO] Epoch 24/25 - Average loss: 0.006309
[INFO] Epoch 25/25 - Average loss: 0.006055
[SUCCESS] Autoencoder training complete.
```

[INFO] Max reconstruction error per sample: tensor([0.0416, 0.0943, 0.1373, 0.0756, 0.0736, 0.0260, 0.0220, 0.3777, 0.0852, 0.1365, 0.0824, 0.0751, 0.0995, 0.0260, 0.0449, 0.1380, 0.1671, 0.0485, 0.0589, 0.0822, 0.0347, 0.0449, 0.1380, 0.0445, 0.0980, 0.0434, 0.1380, 0.1183, 0.0736, 0.0936, 0.0451, 0.1380, 0.0548, 0.6911, 0.0971, 0.0557, 0.0826, 0.0260, 0.1334, 0.0955, 0.0210, 0.0432, 0.1380, 0.0489, 0.0596, 0.0874, 0.0550, 0.1374, 0.0836, 0.0962, 0.0570, 0.0221, 0.0439, 0.0862, 0.4349, 0.0953, 0.0943, 0.0914, 0.0586, 0.0942, 0.0765, 0.1375, 0.0795, 0.0397, 0.0736, 0.0195, 0.0467, 0.0952, 0.0947, 0.0282, 0.1007, 0.0687, 0.0761, 0.0428, 0.0508, 0.1380, 0.0260, 0.0302, 0.1050, 0.0736, 0.0523, 0.0726, 0.3566, 0.3200, 0.0214, 0.2142, 0.1331, 0.1372, 0.5635, 0.0563, 0.0981, 0.0541, 0.0298, 0.0506, 0.0574, 0.1344, 0.0818, 0.0569, 0.0953, 0.0475, 0.1380, 0.2204, 0.0260, 0.2144, 0.0503, 0.0593, 0.0925, 0.0922, 0.0866, 0.6963, 0.0321, 0.0257, 0.0414, 0.0260, 0.0260, 0.0311, 0.0466, 0.1380, 0.0539, 0.1249, 0.3968, 0.0260, 0.5508, 0.6917, 0.0568, 0.0334, 0.0389, 0.0569, 0.0859, 0.0260, 0.0493, 0.0302, 0.0950, 0.0946, 0.0516, 0.0286, 0.0969, 0.5522, 0.0671, 0.2019, 0.3848, 0.0144, 0.0982, 0.0260, 0.0939, 0.0283, 0.1019, 0.1450, 0.1294, 0.1345, 0.0753, 0.0479, 0.1048, 0.0613, 0.0726, 0.0370, 0.1340, 0.4624, 0.0629, 0.1051, 0.0495, 0.0593, 0.0935, 0.0763, 0.0611, 0.0947, 0.2006, 0.0898, 0.0216, 0.0748, 0.2236, 0.0951, 0.0949, 0.0949, 0.0232, 0.5856, 0.0175, 0.2813, 0.2010, 0.3249, 0.3511, 0.0687, 0.0299, 0.0955, 0.0260, 0.1368, 0.0109, 0.3291, 0.0861, 0.0543, 0.0942, 0.1267, 0.0589, 0.1383, 0.0826, 0.0646, 0.1028, 0.0590, 0.1296, 0.0800, 0.0508, 0.1380, 0.1353, 0.1757, 0.0951, 0.0598, 0.0760, 0.0687, 0.0946, 0.0309, 0.0687, 0.0857, 0.0584, 0.0689, 0.1043, 0.0523, 0.0726, 0.0941, 0.0687, 0.1020, 0.0373, 0.0553, 0.0257, 0.2184, 0.0756, 0.0923, 0.0953, 0.3200, 0.3571, 0.0294, 0.0726, 0.1009, 0.2236, 0.0984, 0.0926, 0.1550, 0.2086, 0.1332, 0.0949, 0.0311, 0.0361, 0.0257, 0.1303, 0.0815, 0.0391, 0.1384, 0.0523, 0.0593, 0.2236, 0.0331, 0.4299, 0.0870, 0.1263, 0.0687, 0.0523, 0.0380, 0.0321, 0.0257, 0.2236, 0.0938, 0.2236, 0.1315, 0.0330, 0.0723, 0.0881, 0.0629, 0.0933, 0.1358, 0.0467, 0.3853, 0.0382, 0.1209, 0.1289, 0.1064, 0.1179, 0.1061, 0.0296, 0.0978, 0.2236, 0.1027, 0.0181, 0.0417, 0.0523, 0.1307, 0.0751, 0.3211, 0.0935, 0.1212, 0.1335, 0.1179, 0.1208, 0.1419, 0.1179, 0.0249, 0.0726, 0.5232, 0.1300, 0.0756, 0.0954, 0.0469, 0.0422, 0.0716, 0.0257, 0.0276, 0.0769, 0.2236, 0.0545, 0.2236, 0.0424, 0.0997, 0.0941, 0.1437, 0.0949, 0.0474, 0.0598,

```
[INFO] Number of well-explained examples: 204/822
[INFO] Iteration 2/3 for SymRef(Translate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=204
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=9, hidden_dim=8
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.187930
[INFO] Epoch 2/25 - Average loss: 0.174647
[INFO] Epoch 3/25 - Average loss: 0.162704
[INFO] Epoch 4/25 - Average loss: 0.151336
[INFO] Epoch 5/25 - Average loss: 0.140172
[INFO] Epoch 6/25 - Average loss: 0.128742
[INFO] Epoch 7/25 - Average loss: 0.116829
[INFO] Epoch 8/25 - Average loss: 0.104964
[INFO] Epoch 9/25 - Average loss: 0.092708
[INFO] Epoch 10/25 - Average loss: 0.080672
[INFO] Epoch 11/25 - Average loss: 0.068909
[INFO] Epoch 12/25 - Average loss: 0.057833
[INFO] Epoch 13/25 - Average loss: 0.048586
[INFO] Epoch 14/25 - Average loss: 0.042080
[INFO] Epoch 15/25 - Average loss: 0.039039
[INFO] Epoch 16/25 - Average loss: 0.038291
[INFO] Epoch 17/25 - Average loss: 0.037883
[INFO] Epoch 18/25 - Average loss: 0.037083
[INFO] Epoch 19/25 - Average loss: 0.036395
[INFO] Epoch 20/25 - Average loss: 0.035751
[INFO] Epoch 21/25 - Average loss: 0.034835
[INFO] Epoch 22/25 - Average loss: 0.033542
[INFO] Epoch 23/25 - Average loss: 0.032308
[INFO] Epoch 24/25 - Average loss: 0.030749
[INFO] Epoch 25/25 - Average loss: 0.029157
[SUCCESS] Autoencoder training complete.
```

[INFO] Max reconstruction error per sample: tensor([0.2468, 0.2823, 1.0316, 0.3609, 0.2430, 0.2231, 0.2840, 0.7306, 0.2876, 1.0284, 0.4481, 0.4259, 0.9871, 0.2231, 0.4071, 0.9620, 0.2623, 0.3167, 0.4069, 0.2551, 0.1582, 0.4092, 0.9620, 0.1582, 0.3893, 0.4071, 0.9620, 0.2365, 0.2430, 0.2797, 0.4089, 0.9620, 0.2609, 1.0034, 0.3711, 0.4102, 0.1498, 0.2231, 0.4364, 0.2203, 0.1984, 0.4045, 0.9620, 0.2385, 0.4099, 0.2571, 0.2571, 1.0241, 0.3200, 0.2202, 0.2642, 0.6123, 0.6312, 0.6124, 1.0314, 0.2202, 0.3710, 0.2837, 0.2686, 0.3710, 0.2961, 0.3903, 0.4274, 0.3151, 0.2430, 0.1491, 0.3060, 0.1737, 0.2716, 0.4532, 0.2915, 0.3228, 0.1708, 0.5406, 0.4132, 0.9620, 0.2231, 0.4524, 0.2868, 0.2430, 0.6295, 0.2090, 1.0973, 1.1468, 0.2629, 0.3504, 0.4413, 1.0285, 0.7409, 0.3469, 0.9871, 0.2335, 0.1631, 0.3620, 0.2618, 1.0287, 0.4418, 0.3432, 0.9869, 0.4069, 0.9620, 0.3247, 0.2231, 0.3511, 0.3183, 0.4089, 0.2572, 0.2831, 0.2557, 0.9301, 0.4077, 0.2766, 0.1619, 0.2231, 0.2231, 0.4624, 0.4466, 0.9620, 0.2567, 1.0325, 0.7465, 0.2231, 1.0182, 1.0031, 0.1317, 0.1661, 0.2930, 0.3955, 0.2578, 0.2231, 0.2064, 0.4528, 0.2861, 0.2203, 0.3518, 0.4529, 0.2863, 0.7792, 0.2600, 0.3291, 0.7185, 0.1444, 0.3711, 0.2231, 0.2586, 0.4534, 0.2903, 0.3740, 0.2882, 1.0286, 0.4535, 0.4110, 0.9866, 0.2624, 0.2734, 0.4814, 1.0287, 0.7316, 0.3485, 0.9866, 0.3167, 0.4090, 0.2602, 0.2881, 0.2723, 0.2716, 0.3323, 0.9905, 0.2239, 0.3010, 0.4101, 0.1704, 0.1736, 0.1736, 0.2009, 0.7866, 0.3660, 0.4144, 1.0269, 1.0266, 0.7320, 0.3228, 0.4531, 0.2848, 0.2231, 1.0285, 0.2752, 0.6703, 0.9873, 0.2629, 0.2890, 0.3220, 0.2605, 1.0285, 0.4535, 0.3543, 0.9875, 0.2535, 0.3215, 0.1466, 0.4131, 0.9620, 0.2037, 0.2716, 0.1705, 0.4435, 0.2952, 0.3228, 0.1736, 0.2776, 0.3228, 0.2189, 0.1044, 0.2756, 0.3711, 0.6295, 0.2090, 0.1729, 0.3228, 0.4416, 0.1541, 0.3754, 0.2766, 0.3562, 0.2730, 0.1553, 0.3460, 1.1468, 1.0970, 0.3523, 0.2090, 0.3711, 0.4101, 0.2500, 0.2190, 0.2269, 0.3437, 0.4387, 0.1736, 0.3959, 0.4267, 0.2766, 1.0575, 0.3505, 0.5624, 0.3595, 0.6295, 0.2848, 0.4101, 0.1574, 0.6830, 0.2580, 0.3222, 0.3228, 0.6295, 0.4299, 0.4077, 0.2766, 0.4101, 0.1728, 0.4101, 1.0287, 0.2691, 0.4328, 0.9867, 0.4425, 0.2169, 0.2045, 0.3060, 0.6507, 0.2116, 0.4012, 1.0290, 0.1995, 0.3750, 0.1825, 0.4534, 0.2893, 0.4101, 0.1731, 0.2187, 0.2432, 0.6295, 1.0288, 0.4532, 0.6649, 0.9874, 0.3921, 1.0289, 0.3750, 0.3907, 1.0290, 0.3750, 0.2626, 0.4419, 0.7560, 1.0287, 0.4313, 0.9868, 0.2220, 0.4982, 0.2707, 0.2766, 0.1312, 0.2924, 0.4101, 0.2382, 0.4101, 0.1576, 0.3995, 0.1729, 1.0261, 0.3872, 0.2280, 0.6448

```
[INFO] Number of well-explained examples: 1/822
[INFO] Iteration 3/3 for SymRef(Translate)
[INFO] Preparing DataLoader: batch_size=64, mask sum=1
[SUCCESS] DataLoader prepared successfully.
[INFO] Initializing Autoencoder: input_dim=9, hidden_dim=8
[SUCCESS] Autoencoder initialized successfully.
[INFO] Training autoencoder for 25 epochs, learning rate=0.001
[INFO] Epoch 1/25 - Average loss: 0.175781
[INFO] Epoch 2/25 - Average loss: 0.172784
[INFO] Epoch 3/25 - Average loss: 0.169811
[INFO] Epoch 4/25 - Average loss: 0.166862
[INFO] Epoch 5/25 - Average loss: 0.163935
[INFO] Epoch 6/25 - Average loss: 0.161036
[INFO] Epoch 7/25 - Average loss: 0.158163
[INFO] Epoch 8/25 - Average loss: 0.155398
[INFO] Epoch 9/25 - Average loss: 0.152638
[INFO] Epoch 10/25 - Average loss: 0.149880
[INFO] Epoch 11/25 - Average loss: 0.147124
[INFO] Epoch 12/25 - Average loss: 0.144368
[INFO] Epoch 13/25 - Average loss: 0.141612
[INFO] Epoch 14/25 - Average loss: 0.138853
[INFO] Epoch 15/25 - Average loss: 0.136090
[INFO] Epoch 16/25 - Average loss: 0.133320
[INFO] Epoch 17/25 - Average loss: 0.130527
[INFO] Epoch 18/25 - Average loss: 0.127711
[INFO] Epoch 19/25 - Average loss: 0.124918
[INFO] Epoch 20/25 - Average loss: 0.122133
[INFO] Epoch 21/25 - Average loss: 0.119335
[INFO] Epoch 22/25 - Average loss: 0.116522
[INFO] Epoch 23/25 - Average loss: 0.113693
[INFO] Epoch 24/25 - Average loss: 0.110928
[INFO] Epoch 25/25 - Average loss: 0.108152
[SUCCESS] Autoencoder training complete.
```

[INFO] Max reconstruction error per sample: tensor([0.9035, 0.9029, 1.0388, 0.9036, 0.9026, 0.9037, 0.9058, 0.9032, 0.9025, 1.0531, 0.9032, 0.9028, 0.8966, 0.9037, 0.9031, 0.8920, 0.9027, 0.8990, 0.9023, 0.9031, 0.9046, 0.9030, 0.8920, 0.9048, 0.9018, 0.9029, 0.8920, 0.9040, 0.9026, 0.9029, 0.9031, 0.8920, 0.9037, 0.9050, 0.9023, 0.9027, 0.9044, 0.9037, 0.8987, 0.9034, 0.9039, 0.9030, 0.8920, 0.9035, 0.9022, 0.9031, 0.9038, 1.0389, 0.9038, 0.9034, 0.9037, 0.8980, 0.8974, 0.8983, 1.0512, 0.9034, 0.9023, 0.9031, 0.9037, 0.9023, 0.8999, 0.9033, 0.9022, 0.9028, 0.9026, 0.9050, 0.8999, 0.9046, 0.9033, 0.9053, 0.9023, 0.9025, 0.9038, 0.8999, 0.9030, 0.8920, 0.9037, 0.9055, 0.9025, 0.9026, 0.8973, 0.9036, 1.0541, 1.0565, 0.9036, 0.9038, 0.8987, 1.0531, 0.9045, 0.9026, 0.8966, 0.9035, 0.9048, 0.8989, 0.9037, 1.0528, 0.9032, 0.9031, 0.8966, 0.9032, 0.8920, 0.9014, 0.9037, 0.9039, 0.8990, 0.9023, 0.9030, 0.9031, 0.9031, 0.9054, 0.8988, 0.9001, 0.9049, 0.9037, 0.9037, 0.9055, 0.9023, 0.8920, 0.9037, 1.0410, 0.9042, 0.9037, 1.0522, 0.9051, 0.9039, 0.9047, 0.8993, 0.9023, 0.9031, 0.9037, 0.9052, 0.9056, 0.9030, 0.9033, 0.8986, 0.9053, 0.9031, 0.9046, 0.9016, 0.9036, 0.9041, 0.9049, 0.9023, 0.9037, 0.9029, 0.9054, 0.9029, 0.8982, 0.9034, 1.0531, 0.9032, 0.9026, 0.8965, 0.9033, 0.9013, 0.9043, 1.0531, 0.9010, 0.9032, 0.8966, 0.8991, 0.9022, 0.9030, 0.9001, 0.9031, 0.9033, 0.9041, 0.8986, 0.9038, 0.8996, 0.9037, 0.9043, 0.9045, 0.9045, 0.9040, 0.9056, 0.8998, 0.9036, 1.0540, 1.0520, 0.9014, 0.9025, 0.9057, 0.9032, 0.9037, 1.0530, 0.9001, 0.9036, 0.8964, 0.9033, 0.9031, 0.9024, 0.9028, 1.0531, 0.9033, 0.9030, 0.8966, 0.9032, 0.9018, 0.9019, 0.9030, 0.8920, 0.9028, 0.9029, 0.9043, 0.9013, 0.8997, 0.9025, 0.9044, 0.9056, 0.9025, 0.9038, 0.9039, 0.9022, 0.9023, 0.8973, 0.9036, 0.9046, 0.9025, 0.9022, 0.9047, 0.8990, 0.9001, 0.9037, 0.9050, 0.9023, 0.9021, 1.0565, 1.0541, 0.8993, 0.9036, 0.9023, 0.9037, 0.8996, 0.9035, 0.9023, 0.9039, 0.8987, 0.9045, 0.8989, 0.9036, 0.9001, 1.0389, 0.9031, 0.9021, 0.9023, 0.8973, 0.9031, 0.9037, 0.9045, 0.9004, 0.9031, 0.9019, 0.9025, 0.8973, 0.9039, 0.8988, 0.9001, 0.9037, 0.9045, 0.9037, 1.0530, 0.8998, 0.9028, 0.8966, 0.9007, 0.9034, 0.9028, 0.8999, 0.9037, 0.9051, 0.9003, 1.0522, 0.9013, 0.9039, 0.9033, 0.9057, 0.9030, 0.9037, 0.9023, 0.9037, 0.9034, 0.8973, 1.0529, 0.9034, 0.9035, 0.8965, 0.9003, 1.0524, 0.9039, 0.9004, 1.0520, 0.9039, 0.9060, 0.9044, 0.9030, 1.0528, 0.9028, 0.8966, 0.9044, 0.9011, 0.9023, 0.9001, 0.9044, 0.9000, 0.9037, 0.9034, 0.9037, 0.9048, 0.9019, 0.9046, 1.0377, 0.9032, 0.9038, 0.9019,

```

[INFO] Number of well-explained examples: 0/822
[SUCCESS] Model training finished for SymRef(Translate). Total well-
explained examples: 0/822
[INFO] Skipping SymTrans(Translate) (only 42 examples)
[INFO] Skipping Union(SymTrans) (only 42 examples)
[INFO] Skipping Union(SymRot) (only 23 examples)
[INFO] Skipping SymRot(Union) (only 14 examples)
[INFO] Skipping SymRot(Translate) (only 9 examples)
[SUCCESS] Finished training 7 PAIRS
[SUCCESS] Training complete. 4 singleton models and 7 pair models
trained.

```

```

[65]: # Test on a random DSL shape
if all_dsl_shapes and singleton_models and pair_models:
    random_chair = random.choice(all_dsl_shapes)
    debug_info("--- ORIGINAL CHAIR ---")
    debug_info(f"Type: {type(random_chair).__name__}")
    debug_info(f"Serialized children count: {len(random_chair.
↳serialize()[1][1])}")
    debug_info(f"Preview: {random_chair}")

    # Integrate abstractions
    debug_info("Integrating abstractions...")
    abstracted_chair = integrate_abstractions(
        random_chair, singleton_models, pair_models, error_threshold=0.01
    )

    debug_info("\n--- ABSTRACTED CHAIR ---")
    debug_info(f"Type: {type(abstracted_chair).__name__}")
    if isinstance(abstracted_chair, Abstraction):
        debug_info(f"Abstraction pattern: {abstracted_chair.pattern_name},
↳compressed dim: {len(abstracted_chair.compressed_params)}")
        debug_info(f"Preview: {abstracted_chair}")

    # Visualization
    try:
        debug_info("Plotting original and abstracted DSL shapes...")
        plot_dsl_with_k3d(random_chair)
        plot_dsl_with_k3d(abstracted_chair)
        debug_success("Visualization complete.")
    except Exception as e:
        debug_error("Plotting failed:", e)
else:
    debug_error("Cannot run test: DSL shapes or trained models are missing.")

```

```

[INFO] --- ORIGINAL CHAIR ---
[INFO] Type: Union

```

```
[INFO] Serialized children count: 2
```

[INFO] Preview: Union(

```
Union(
  Translate(center=[-0.005, -0.290, 0.092])
    Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
      Scale(lengths=[0.764, 0.051, 0.764])
        Box(label=1),
  SymRef(
    plane=[1.000, -0.000, -0.000],
    point_on_plane=[-0.004, -0.566, -0.117]
  )(
    Union(
      Translate(center=[-0.346, -0.566, -0.117])
        Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
          Scale(lengths=[0.077, 0.503, 0.077])
            Box(label=2),
      Translate(center=[-0.346, -0.566, 0.434])
        Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
          Scale(lengths=[0.077, 0.503, 0.077])
            Box(label=2)
    )
  )
),
Union(
  Union(
    Translate(center=[-0.004, 0.722, -0.138])
      Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
        Scale(lengths=[0.762, 0.222, 0.274])
          Box(label=0),
    Translate(center=[-0.001, 0.261, -0.206])
      Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
        Scale(lengths=[0.058, 1.054, 0.058])
          Box(label=0)
  ),
  SymRef(
    plane=[1.000, 0.000, 0.005],
    point_on_plane=[-0.002, 0.261, -0.127]
```

```

[INFO] Integrating abstractions...
[INFO]       [Rebuilt] Box with 1 children
[INFO]       [RETURN NODE] Box
[INFO]       [Rebuilt] Scale with 1 children
[INFO]       [PAIR CHECK] Scale(Box), error=0.1339
[INFO]       [SINGLETON CHECK] Scale, error=0.0889
[INFO]       [RETURN NODE] Scale
[INFO]       [Rebuilt] Rotate with 1 children
[INFO]       [PAIR CHECK] Rotate(Scale), error=0.0034
[SUCCESS]       [PAIR ABSTRACTION CREATED] Rotate(Scale)
[INFO]       [Rebuilt] Translate with 1 children
[INFO]       [SINGLETON CHECK] Translate, error=0.2169
[INFO]       [RETURN NODE] Translate
[INFO]       [Rebuilt] Box with 1 children
[INFO]       [RETURN NODE] Box
[INFO]       [Rebuilt] Scale with 1 children
[INFO]       [PAIR CHECK] Scale(Box), error=0.0144
[INFO]       [SINGLETON CHECK] Scale, error=0.0108
[INFO]       [RETURN NODE] Scale
[INFO]       [Rebuilt] Rotate with 1 children
[INFO]       [PAIR CHECK] Rotate(Scale), error=0.0014
[SUCCESS]       [PAIR ABSTRACTION CREATED] Rotate(Scale)
[INFO]       [Rebuilt] Translate with 1 children
[INFO]       [SINGLETON CHECK] Translate, error=0.0631
[INFO]       [RETURN NODE] Translate
[INFO]       [Rebuilt] Box with 1 children
[INFO]       [RETURN NODE] Box
[INFO]       [Rebuilt] Scale with 1 children
[INFO]       [PAIR CHECK] Scale(Box), error=0.0144
[INFO]       [SINGLETON CHECK] Scale, error=0.0108
[INFO]       [RETURN NODE] Scale
[INFO]       [Rebuilt] Rotate with 1 children
[INFO]       [PAIR CHECK] Rotate(Scale), error=0.0014
[SUCCESS]       [PAIR ABSTRACTION CREATED] Rotate(Scale)
[INFO]       [Rebuilt] Translate with 1 children
[INFO]       [SINGLETON CHECK] Translate, error=0.0606
[INFO]       [RETURN NODE] Translate
[INFO]       [Rebuilt] Union with 2 children
[INFO]       [RETURN NODE] Union
[INFO]       [Rebuilt] SymRef with 1 children
[INFO]       [PAIR CHECK] SymRef(Union), error=0.0131
[INFO]       [SINGLETON CHECK] SymRef, error=0.0080
[SUCCESS]       [SINGLETON ABSTRACTION CREATED] SymRef
[INFO]       [Rebuilt] Union with 2 children
[INFO]       [PAIR CHECK] Union(Translate), error=0.0937
[INFO]       [RETURN NODE] Union
[INFO]       [Rebuilt] Box with 1 children
[INFO]       [RETURN NODE] Box

```

```

[INFO]           [Rebuilt] Scale with 1 children
[INFO]           [PAIR CHECK] Scale(Box), error=0.1756
[INFO]           [SINGLETON CHECK] Scale, error=0.0872
[INFO]           [RETURN NODE] Scale
[INFO]           [Rebuilt] Rotate with 1 children
[INFO]           [PAIR CHECK] Rotate(Scale), error=0.0044
[SUCCESS]        [PAIR ABSTRACTION CREATED] Rotate(Scale)
[INFO]           [Rebuilt] Translate with 1 children
[INFO]           [SINGLETON CHECK] Translate, error=0.0794
[INFO]           [RETURN NODE] Translate
[INFO]           [Rebuilt] Box with 1 children
[INFO]           [RETURN NODE] Box
[INFO]           [Rebuilt] Scale with 1 children
[INFO]           [PAIR CHECK] Scale(Box), error=0.0106
[INFO]           [SINGLETON CHECK] Scale, error=0.0110
[INFO]           [RETURN NODE] Scale
[INFO]           [Rebuilt] Rotate with 1 children
[INFO]           [PAIR CHECK] Rotate(Scale), error=0.0024
[SUCCESS]        [PAIR ABSTRACTION CREATED] Rotate(Scale)
[INFO]           [Rebuilt] Translate with 1 children
[INFO]           [SINGLETON CHECK] Translate, error=0.0255
[INFO]           [RETURN NODE] Translate
[INFO]           [Rebuilt] Union with 2 children
[INFO]           [RETURN NODE] Union
[INFO]           [Rebuilt] Box with 1 children
[INFO]           [RETURN NODE] Box
[INFO]           [Rebuilt] Scale with 1 children
[INFO]           [PAIR CHECK] Scale(Box), error=0.0106
[INFO]           [SINGLETON CHECK] Scale, error=0.0110
[INFO]           [RETURN NODE] Scale
[INFO]           [Rebuilt] Rotate with 1 children
[INFO]           [PAIR CHECK] Rotate(Scale), error=0.0024
[SUCCESS]        [PAIR ABSTRACTION CREATED] Rotate(Scale)
[INFO]           [Rebuilt] Translate with 1 children
[INFO]           [SINGLETON CHECK] Translate, error=0.0204
[INFO]           [RETURN NODE] Translate
[INFO]           [Rebuilt] Box with 1 children
[INFO]           [RETURN NODE] Box
[INFO]           [Rebuilt] Scale with 1 children
[INFO]           [PAIR CHECK] Scale(Box), error=0.0106
[INFO]           [SINGLETON CHECK] Scale, error=0.0110
[INFO]           [RETURN NODE] Scale
[INFO]           [Rebuilt] Rotate with 1 children
[INFO]           [PAIR CHECK] Rotate(Scale), error=0.0024
[SUCCESS]        [PAIR ABSTRACTION CREATED] Rotate(Scale)
[INFO]           [Rebuilt] Translate with 1 children
[INFO]           [SINGLETON CHECK] Translate, error=0.0218
[INFO]           [RETURN NODE] Translate

```

```
[INFO]      [Rebuilt] Union with 2 children
[INFO]      [RETURN NODE] Union
[INFO]      [Rebuilt] SymRef with 1 children
[INFO]      [PAIR CHECK] SymRef(Union), error=0.0121
[INFO]      [SINGLETON CHECK] SymRef, error=0.0052
[SUCCESS]    [SINGLETON ABSTRACTION CREATED] SymRef
[INFO]      [Rebuilt] Union with 2 children
[INFO]      [RETURN NODE] Union
[INFO]      [Rebuilt] Union with 2 children
[INFO]      [RETURN NODE] Union
[INFO]
--- ABSTRACTED CHAIR ---
[INFO] Type: Union
```

[INFO] Preview: Union(

```
Union(
  Translate(center=[-0.005, -0.290, 0.092])
    Abs(Rotate(Scale), dim=6)(
      Box(label=1)
    ),
  Abs(SymRef, dim=5)(
    Union(
      Translate(center=[-0.346, -0.566, -0.117])
        Abs(Rotate(Scale), dim=6)(
          Box(label=2)
        ),
      Translate(center=[-0.346, -0.566, 0.434])
        Abs(Rotate(Scale), dim=6)(
          Box(label=2)
        )
    )
  )
),
Union(
  Union(
    Translate(center=[-0.004, 0.722, -0.138])
      Abs(Rotate(Scale), dim=6)(
        Box(label=0)
      ),
    Translate(center=[-0.001, 0.261, -0.206])
      Abs(Rotate(Scale), dim=6)(
        Box(label=0)
      )
  ),
  Abs(SymRef, dim=5)(
    Union(
      Translate(center=[-0.320, 0.261, -0.129])
        Abs(Rotate(Scale), dim=6)(
          Box(label=0)
        )
      )
  )
)
```

```

[INFO] Plotting original and abstracted DSL shapes...
Expanding DSL tree for visualization...
Found 11 total boxes after expansion.

Output()

Expanding DSL tree for visualization...
[INFO] Expanding abstraction: Abs(Rotate(Scale), dim=6)(
    Box(label=1)
)
[INFO] Compressed params tensor shape: (1, 6)
[SUCCESS] Reconstructed params: [-0.0015942193567752838,
0.0033435821533203125, 0.0010694041848182678, 1.0034091472625732,
0.7643609046936035, 0.050254080444574356, 0.7649176120758057]
[INFO] [INSTANTIATE] Pattern: 'Rotate(Scale)' |
params=[-0.0015942193567752838, 0.0033435821533203125, 0.0010694041848182678,
1.0034091472625732, 0.7643609046936035, 0.050254080444574356,
0.7649176120758057] | #children=1
[SUCCESS] Successfully rebuilt node: Rotate(quat=[-0.0016, 0.0033, 0.0011,
1.0034])
    Scale(lengths=[0.764, 0.050, 0.765])
    Box(label=1)
[INFO] Expanding abstraction: Abs(SymRef, dim=5)(
    Union(
        Translate(center=[-0.346, -0.566, -0.117])
        Abs(Rotate(Scale), dim=6)(
            Box(label=2)
        ),
        Translate(center=[-0.346, -0.566, 0.434])
        Abs(Rotate(Scale), dim=6)(
            Box(label=2)
        )
    )
)
[INFO] Compressed params tensor shape: (1, 5)

```

```

[SUCCESS] Reconstructed params: [0.9980296492576599, 0.002478860318660736,
0.0006636390462517738, 0.0036224573850631714, -0.5720623135566711,
-0.11461415141820908]
[INFO] [INSTANTIATE] Pattern: 'SymRef' | params=[0.9980296492576599,
0.002478860318660736, 0.0006636390462517738, 0.0036224573850631714,
-0.5720623135566711, -0.11461415141820908] | #children=1
[INFO] [SINGLETON] SymRef with params length 6
[SUCCESS] Successfully rebuilt node: SymRef(
    plane=[0.998, 0.002, 0.001],
    point_on_plane=[0.004, -0.572, -0.115]
)(
    Union(
        Translate(center=[-0.346, -0.566, -0.117])
        Abs(Rotate(Scale), dim=6)(
            Box(label=2)
        ),
        Translate(center=[-0.346, -0.566, 0.434])
        Abs(Rotate(Scale), dim=6)(
            Box(label=2)
        )
    )
)
[INFO] Expanding abstraction: Abs(Rotate(Scale), dim=6)(
    Box(label=2)
)
[INFO] Compressed params tensor shape: (1, 6)
[SUCCESS] Reconstructed params: [-0.000655006617307663,
0.0013978928327560425, -0.0007073655724525452, 1.0011658668518066,
0.07751844078302383, 0.502310037612915, 0.07814215123653412]
[INFO] [INSTANTIATE] Pattern: 'Rotate(Scale)' |
params=[-0.000655006617307663, 0.0013978928327560425, -0.0007073655724525452,
1.0011658668518066, 0.07751844078302383, 0.502310037612915, 0.07814215123653412]
| #children=1

```

```

[SUCCESS] Successfully rebuilt node: Rotate(quat=[-0.0007, 0.0014, -0.0007,
1.0012])
    Scale(lengths=[0.078, 0.502, 0.078])
        Box(label=2)
[INFO] Expanding abstraction: Abs(Rotate(Scale), dim=6)(
    Box(label=2)
)
[INFO] Compressed params tensor shape: (1, 6)
[SUCCESS] Reconstructed params: [-0.000655006617307663,
0.0013978928327560425, -0.0007073655724525452, 1.0011658668518066,
0.07751844078302383, 0.502310037612915, 0.07814215123653412]
[INFO] [INSTANTIATE] Pattern: 'Rotate(Scale)' |
params=[-0.000655006617307663, 0.0013978928327560425, -0.0007073655724525452,
1.0011658668518066, 0.07751844078302383, 0.502310037612915, 0.07814215123653412]
| #children=1
[SUCCESS] Successfully rebuilt node: Rotate(quat=[-0.0007, 0.0014, -0.0007,
1.0012])
    Scale(lengths=[0.078, 0.502, 0.078])
        Box(label=2)
[INFO] Expanding abstraction: Abs(Rotate(Scale), dim=6)(
    Box(label=0)
)
[INFO] Compressed params tensor shape: (1, 6)
[SUCCESS] Reconstructed params: [-0.0024891607463359833,
0.004352003335952759, 0.00045206397771835327, 1.0024285316467285,
0.7623201012611389, 0.2219916433095932, 0.27538442611694336]
[INFO] [INSTANTIATE] Pattern: 'Rotate(Scale)' |
params=[-0.0024891607463359833, 0.004352003335952759, 0.00045206397771835327,
1.0024285316467285, 0.7623201012611389, 0.2219916433095932, 0.27538442611694336]
| #children=1
[SUCCESS] Successfully rebuilt node: Rotate(quat=[-0.0025, 0.0044, 0.0005,
1.0024])
    Scale(lengths=[0.762, 0.222, 0.275])
        Box(label=0)

```

```

[INFO] Expanding abstraction: Abs(Rotate(Scale), dim=6)(
    Box(label=0)
)
[INFO] Compressed params tensor shape: (1, 6)
[SUCCESS] Reconstructed params: [-0.002308238297700882,
0.0023776888847351074, -0.0020132139325141907, 1.0011844635009766,
0.058474861085414886, 1.0539015531539917, 0.05903676152229309]
[INFO] [INSTANTIATE] Pattern: 'Rotate(Scale)' |
params=[-0.002308238297700882, 0.0023776888847351074, -0.0020132139325141907,
1.0011844635009766, 0.058474861085414886, 1.0539015531539917,
0.05903676152229309] | #children=1
[SUCCESS] Successfully rebuilt node: Rotate(quat=[-0.0023, 0.0024, -0.0020,
1.0012])
    Scale(lengths=[0.058, 1.054, 0.059])
    Box(label=0)
[INFO] Expanding abstraction: Abs(SymRef, dim=5)(
    Union(
        Translate(center=[-0.320, 0.261, -0.129])
            Abs(Rotate(Scale), dim=6)(
                Box(label=0)
            ),
        Translate(center=[-0.165, 0.261, -0.187])
            Abs(Rotate(Scale), dim=6)(
                Box(label=0)
            )
    )
)
[INFO] Compressed params tensor shape: (1, 5)
[SUCCESS] Reconstructed params: [0.9992480874061584, 0.001408994197845459,
-6.382446736097336e-05, 0.0027413219213485718, 0.26635414361953735,
-0.1262945830821991]
[INFO] [INSTANTIATE] Pattern: 'SymRef' | params=[0.9992480874061584,
0.001408994197845459, -6.382446736097336e-05, 0.0027413219213485718,
0.26635414361953735, -0.1262945830821991] | #children=1
[INFO] [SINGLETON] SymRef with params length 6

```

```

[SUCCESS] Successfully rebuilt node: SymRef(
  plane=[0.999, 0.001, -0.000],
  point_on_plane=[0.003, 0.266, -0.126]
)(
  Union(
    Translate(center=[-0.320, 0.261, -0.129])
      Abs(Rotate(Scale), dim=6)(
        Box(label=0)
      ),
    Translate(center=[-0.165, 0.261, -0.187])
      Abs(Rotate(Scale), dim=6)(
        Box(label=0)
      )
  )
)
[INFO] Expanding abstraction: Abs(Rotate(Scale), dim=6)(
  Box(label=0)
)
[INFO] Compressed params tensor shape: (1, 6)
[SUCCESS] Reconstructed params: [-0.0023078210651874542,
0.002377241849899292, -0.0020133107900619507, 1.0011842250823975,
0.058363787829875946, 1.0539014339447021, 0.05892661213874817]
[INFO] [INSTANTIATE] Pattern: 'Rotate(Scale)' |
params=[-0.0023078210651874542, 0.002377241849899292, -0.0020133107900619507,
1.0011842250823975, 0.058363787829875946, 1.0539014339447021,
0.05892661213874817] | #children=1
[SUCCESS] Successfully rebuilt node: Rotate(quat=[-0.0023, 0.0024, -0.0020,
1.0012])
  Scale(lengths=[0.058, 1.054, 0.059])
  Box(label=0)
[INFO] Expanding abstraction: Abs(Rotate(Scale), dim=6)(
  Box(label=0)
)
[INFO] Compressed params tensor shape: (1, 6)

```

```

[SUCCESS] Reconstructed params: [-0.002308044582605362,
0.0023775100708007812, -0.0020132511854171753, 1.0011844635009766,
0.05842875689268112, 1.0539015531539917, 0.05899176001548767]
[INFO] [INSTANTIATE] Pattern: 'Rotate(Scale)' |
params=[-0.002308044582605362, 0.0023775100708007812, -0.0020132511854171753,
1.0011844635009766, 0.05842875689268112, 1.0539015531539917,
0.05899176001548767] | #children=1
[SUCCESS] Successfully rebuilt node: Rotate(quat=[-0.0023, 0.0024, -0.0020,
1.0012])
    Scale(lengths=[0.058, 1.054, 0.059])
    Box(label=0)

```

Found 11 total boxes after expansion.

Output()

[SUCCESS] Visualization complete.

```
[61]: print(abstracted_chair)
```

```

Union(
  Union(
    Union(
      Translate(center=[-0.001, -0.111, 0.158])
        Abs(Rotate(Scale), dim=6)(
          Box(label=1)
        ),
      SymRef(
        plane=[1.000, 0.027, 0.001],
        point_on_plane=[-0.005, 0.071, 0.499]
      )(
        Union(
          Translate(center=[-0.417, 0.060, 0.498])
            Rotate(quat=[0.0121, 0.0158, 0.0684, 0.9975])
              Scale(lengths=[0.080, 0.364, 0.079])
                Box(label=3),
          Translate(center=[-0.413, 0.019, 0.079])
            Rotate(quat=[-0.0000, 0.0000, 0.0606, 0.9982])
              Scale(lengths=[0.093, 0.459, 0.794])
                Box(label=3)
          )
        )
      ),
    Abs(SymRef, dim=5)(
      Union(
        Abs(Translate, dim=2),

```

```

        Translate(center=[-0.403, -0.421, 0.499])
        Abs(Rotate, dim=3)
    )
),
Translate(center=[-0.003, 0.173, -0.248])
    Rotate(quat=[-0.0886, -0.0248, -0.0036, 0.9958])
    Scale(lengths=[0.945, 0.800, 0.294])
    Box(label=0)
)

```

```
[62]: print(random_chair)
```

```

Union(
    Union(
        Union(
            Translate(center=[-0.001, -0.111, 0.158])
            Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
            Scale(lengths=[0.741, 0.225, 0.746])
            Box(label=1),
            SymRef(
                plane=[1.000, 0.027, 0.001],
                point_on_plane=[-0.005, 0.071, 0.499]
            )(
                Union(
                    Translate(center=[-0.417, 0.060, 0.498])
                    Rotate(quat=[0.0121, 0.0158, 0.0684, 0.9975])
                    Scale(lengths=[0.080, 0.364, 0.079])
                    Box(label=3),
                    Translate(center=[-0.413, 0.019, 0.079])
                    Rotate(quat=[-0.0000, 0.0000, 0.0606, 0.9982])
                    Scale(lengths=[0.093, 0.459, 0.794])
                    Box(label=3)
                )
            )
        ),
        SymRef(
            plane=[1.000, -0.000, 0.000],
            point_on_plane=[-0.005, -0.252, -0.280]
        )(
            Union(
                Translate(center=[-0.414, -0.252, -0.280])
                Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
                Scale(lengths=[0.093, 0.988, 0.072])
                Box(label=2),
                Translate(center=[-0.403, -0.421, 0.499])
                Rotate(quat=[0.0000, 0.0000, 0.0000, 1.0000])
                Scale(lengths=[0.070, 0.650, 0.072])
            )
        )
    )
)

```

```

        Box(label=2)
    )
)
),
Translate(center=[-0.003, 0.173, -0.248])
    Rotate(quat=[-0.0886, -0.0248, -0.0036, 0.9958])
        Scale(lengths=[0.945, 0.800, 0.294])
            Box(label=0)
)

```

[]: